

**DESIGN AND IMPLEMENTATION OF YOBE STATE E-TRANSPORT
BOOKING AND MANAGEMENT SYSTEM**

BY

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SFE/23U/448

JULY,2026

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**PROJECT SUBMITTED TO THE DEPARTMENT OF SOFTWARE
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FOR THE AWARD OF BACHELOR OF SCIENCE DEGREE IN SOFTWARE
ENGINEERING**

JULY, 2026

DECLARATION

I declare that the work described in this project is original, and has not been previously submitted to any University or similar institutions for the award of any degree or certificate.

Name of Candidate: ABBA MUSTAPHA KACHALLA

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CERTIFICATION

We the undersigned, hereby certify that this project presented by ABBA MUSTAPHA KACHALLA (Matric No. SFE/23U/448) be accepted as fulfilling part of requirements for the award of degree of Bachelor of Science in Software Engineering.

Title: DESIGN AND IMPLEMENTATION OF YOBE STATE E-TRANSPORT BOOKING AND MANAGEMENT SYSTEM

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DEDICATION

This project work is dedicated to God Almighty the most faithful for his endless blessings and protections.

To my father, Kachalla Mustapha Kolomi. Thank you for being the cornerstone of my ambition. Your unwavering support, wisdom, and the sacrifices you made to ensure I received the best education have been my greatest motivation. This achievement is as much yours as it is mine.

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ABSTRACT

Public road transport in Yobe State, Nigeria, continues to rely heavily on manual ticket booking, paper-based records, and fragmented coordination between transport operators and passengers. These practices contribute to long queues at terminals, seat allocation errors, limited booking transparency, and difficulties in monitoring fleet operations. This study addresses these challenges by designing and implementing the Yobe State E-Transport Booking and Management System, a web-based platform that digitizes passenger booking, seat selection, payment, and administrative transport management. The principal aim of this research is to develop a secure, user-friendly, and scalable e-transport system tailored to interstate and intrastate routes originating from Yobe State. The methodology adopted involved requirements analysis, system modelling using Unified Modelling Language (UML) diagrams including use case, activity, and entity-relationship diagrams, relational database design, and iterative prototyping for system implementation. The system was implemented using Next.js with TypeScript for the frontend and server-side application logic, Supabase (PostgreSQL) for database management and authentication, Tailwind CSS for responsive interface design, and integrated services for route mapping, electronic ticketing with QR codes, email notification, and online payment processing (Paystack Payments Limited, 2024). Development was carried out using Visual Studio Code as the integrated development environment. Key results demonstrate successful implementation of passenger features such as route search, real-time seat map selection, secure booking, electronic ticket generation in PDF format, and booking history, as well as administrative modules for managing buses, routes, schedules, drivers, bookings, and operational reports. Testing confirmed that the system reduces manual booking errors, improves passenger convenience, and provides operators with centralized data for decision-making. In conclusion, the developed system offers a practical digital solution for modernizing transport services in Yobe State. It is recommended that transport unions and government agencies adopt and scale the platform, integrate mobile money payment channels, and extend coverage to additional routes to improve service delivery for commuters across the region.

Keywords: E-Transport, Online Booking, Seat Management, Web Application, Yobe State, Transport Management System.

CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

Transportation is a critical pillar of economic development and social connectivity in every nation. It facilitates the movement of people, goods, and services, thereby supporting trade, employment, education, healthcare access, and regional integration. In developing countries, road transport remains the dominant mode of passenger movement due to its flexibility, relatively lower infrastructure cost compared to rail and air transport, and its ability to reach both urban centres and rural communities. As populations grow and urbanization increases, the demand for efficient, reliable, and safe transport services continues to rise, placing greater pressure on transport operators and regulatory authorities to modernize their operations.

In Nigeria, the road transport sector plays a vital role in connecting the thirty-six states and the Federal Capital Territory. Millions of Nigerians depend on interstate and intrastate bus services for daily travel, business trips, pilgrimage, education, and medical visits. Transport companies, motor park unions, and private operators coordinate departures from major terminals, assign vehicles to routes, sell tickets, and manage passenger manifests. Despite the sector's importance, many transport operations across the country still rely on manual processes such as handwritten tickets, verbal seat allocation, cash-only payments at counters, and paper-based record keeping. These traditional methods are increasingly inadequate in an era where passengers expect convenience, transparency, and digital access to services.

Yobe State, located in the northeastern region of Nigeria, is strategically connected to several neighbouring states including Bauchi, Borno, Gombe, and Jigawa, as well as major destinations such as Abuja, Kano, Kaduna, Jos, Sokoto, and Zamfara. Road transport from Yobe serves as an essential link for residents travelling for commerce, civil service, education, and inter-state engagements. Motor parks and transport terminals in cities such as Damaturu and other local government areas experience regular passenger traffic, particularly along high-demand corridors. Operators deploy various vehicle types including Toyota buses, Sienna, Sharon, and Golf vehicles to match route distance, road conditions, and passenger volume. However, the management of bookings, schedules,

fleet allocation, and passenger information in the state largely follows conventional manual practices.

The limitations of manual transport management are well documented in both academic literature and industry practice. Passengers often face long waiting times at terminals, uncertainty about seat availability, duplicate bookings, and difficulty verifying ticket authenticity. Transport operators encounter challenges in tracking revenue, monitoring fleet utilization, preventing seat conflicts, and generating accurate reports for business and regulatory purposes. Administrative staff spend considerable time reconciling paper records, while managers lack real-time visibility into booking trends, route performance, and operational bottlenecks. Furthermore, the absence of centralized digital records makes it difficult to enforce route-specific vehicle regulations, manage cancellations, or respond promptly to passenger enquiries.

Against this background, the design and implementation of an E-Transport Booking and Management System for Yobe State becomes both timely and relevant. Such a system seeks to replace fragmented manual workflows with a unified web-based platform that serves passengers and transport administrators through a single digital environment. By enabling online route search, interactive seat selection, secure booking, electronic ticketing, and centralized administrative control, the proposed solution addresses key operational inefficiencies while improving the overall passenger experience. This study therefore situates itself within the broader discourse on ICT-enabled public service delivery and sector-specific software engineering, with particular focus on applying modern web technologies to solve real transport management problems in Yobe State, Nigeria.

1.2 Statement of the Problem

Despite the growing importance of road transport in Yobe State and across Nigeria, the prevailing methods of managing passenger bookings and transport operations remain largely manual and inefficient. Transport operators, terminal attendants, and administrative staff depend on paper tickets, verbal communication, and physical presence at motor parks to coordinate departures, assign seats, collect fares, and maintain records. This traditional approach creates a wide range of operational, administrative, and service delivery problems that negatively affect both passengers and transport managers.

One major problem is the lack of a centralized digital platform for booking and managing interstate and intrastate trips originating from Yobe State. Passengers must travel to motor parks or booking offices to enquire about trip availability, compare fares, and secure seats.

Another significant problem is inefficient seat allocation and booking conflict. In manual systems, multiple attendants or booking agents may assign the same seat to different passengers, especially during peak travel periods such as festive seasons, market days, and school resumptions.

Payment and ticketing processes also present serious challenges. Most transactions are conducted in cash at terminals, making reconciliation difficult and exposing operators to errors in change calculation, incomplete record keeping, and limited audit trails. Paper tickets can be lost, damaged, forged, or difficult to verify at boarding points.

Additionally, existing generic transport applications developed for other regions do not adequately address the specific operational context of Yobe State. The state's transport network includes distinct routes to destinations such as Abuja, Jos, Bauchi, Kano, Kaduna, Nasarawa, Niger, Sokoto, and Zamfara, as well as connections within local government areas. Certain routes impose vehicle-type restrictions, for example limiting long-distance trips to approved bus categories. Off-the-shelf booking platforms rarely incorporate such localized rules, administrative controls, and reporting needs required by Yobe-based transport operators and regulatory stakeholders.

1.3 Aim and Objectives of the Study

The aim of this study is to design and implement a web-based E-Transport Booking and Management System that improves the efficiency, reliability, and accessibility of road transport services for passengers and operators in Yobe State, Nigeria.

To achieve this aim, the following specific objectives were set:

1. To examine the existing manual transport booking and management processes in Yobe State and identify their operational limitations.

2. To analyse the functional and non-functional requirements for a digital transport booking and management platform tailored to Yobe State routes and operators and design the system architecture, database structure, user interfaces, and workflow models for passenger booking and administrative management.
3. To develop a web application that enables passengers to search routes, view schedules, select seats, complete bookings, and receive electronic tickets and implement administrative modules for managing transport companies, buses, drivers, routes, schedules, and bookings from a centralized dashboard.
4. To integrate secure authentication, real-time seat availability, electronic ticketing with QR codes, and online payment processing into the system and deploy and test the developed system in order to evaluate its performance, usability, and effectiveness in addressing identified transport management problems.

1.4 The Scope of the Study

The scope of this study defines the boundaries within which the research was conducted and the system was designed, developed, and evaluated. It clarifies what the project covers and what lies outside its immediate focus, thereby ensuring a realistic and achievable implementation within the available time and resources.

This study focuses on the design and implementation of a web-based E-Transport Booking and Management System for road transport operations associated with Yobe State, Nigeria. The system targets both interstate trips from Yobe to major destinations such as Abuja, Jos, Bauchi, Kano, Kaduna, Nasarawa, Niger, Sokoto, and Zamfara, as well as intrastate connections within Yobe local government areas. The platform supports multiple vehicle categories used in the state's transport sector, including Toyota buses, Sienna, Sharon, and Golf vehicles, with route-specific rules applied where necessary.

On the passenger side, the scope includes user registration and authentication, route and schedule search, interactive seat map selection, online booking, payment processing, generation of electronic tickets with QR codes, ticket download, booking history, and access to route information. On the administrative side, the scope covers modules for managing transport companies, buses, drivers, routes, schedules, bookings, ticket verification, and basic operational reports through a secure admin dashboard.

Technically, the study is limited to the development of a responsive web application using Next.js with TypeScript, Supabase as the backend database and authentication service, and supporting tools for mapping, email notification, and payment integration. The system was designed for deployment on a cloud hosting environment and tested under simulated and local operational conditions. Emphasis was placed on functional correctness, data security, usability, and the ability to handle typical booking and management workflows encountered in Yobe State transport operations.

1.5 Significance of the Study

The significance of this study lies in its potential to transform how road transport services are accessed, managed, and monitored in Yobe State through the application of information and communication technology. As digital solutions increasingly shape public and commercial services, a locally designed e-transport platform offers practical benefits to passengers, transport operators, administrators, researchers, and the wider society.

To passengers, the developed system provides greater convenience by allowing route search, seat selection, and ticket purchase from any location with internet access. This reduces the need for repeated visits to motor parks, minimizes waiting time, and improves transparency in fare and schedule information. Electronic tickets with verifiable QR codes enhance boarding confidence and reduce disputes associated with lost or invalid paper tickets. Passengers can also retain digital records of their travel history, which is useful for personal planning, reimbursement, and future reference.

For transport operators and managers, the system offers a centralized tool for coordinating fleet deployment, route scheduling, driver assignment, and booking oversight. Real-time seat management reduces overbooking and revenue leakage, while structured digital records simplify daily reconciliation and reporting. Administrators gain access to booking trends, route performance indicators, and operational summaries that support better planning and resource allocation. These capabilities contribute to improved service quality, accountability, and competitiveness in the transport market.

From a governmental and policy perspective, the study demonstrates how state-level digital infrastructure can support orderly regulation of transport activities, standardization

of booking practices, and improved data collection for planning purposes. Although full integration with regulatory agencies lies beyond the immediate scope of this project, the system provides a foundation upon which future e-government transport initiatives can be built. Such digital platforms may also promote safer and more organized terminal operations by reducing overcrowding and informal booking practices.

1.6 Definition of Terms

For clarity and consistency, the following terms are defined as they are used in this project report:

Administrator: A authorized user responsible for managing transport operations through the system dashboard, including buses, routes, schedules, drivers, bookings, and reports.

Authentication: The process of verifying the identity of a user before granting access to protected features of the system, such as passenger accounts or the admin dashboard.

Booking: The act of reserving one or more seats on a scheduled trip, usually accompanied by passenger details, fare payment, and ticket issuance.

Cloud Computing: The delivery of computing services such as databases, storage, and application hosting over the internet, enabling scalable and remote system deployment.

Database: An organized collection of structured data stored electronically and managed through a database management system, used to persist routes, schedules, bookings, users, and related records.

E-Ticket (Electronic Ticket): A digital travel document generated by the system after a successful booking, containing trip details and a verifiable code or QR image for boarding confirmation.

E-Transport: The use of electronic and web-based technologies to support transport-related services such as trip search, booking, payment, ticketing, and operational management.

Fleet Management: The administrative control of a group of transport vehicles, including registration, assignment to routes, capacity tracking, and operational monitoring.

Interstate Transport: Road passenger movement between Yobe State and other states in Nigeria, such as trips to Abuja, Kano, Kaduna, and Jos.

Intrastate Transport: Road passenger movement within Yobe State, including connections between local government areas and trips that do not cross state boundaries.

Local Government Area (LGA): An administrative subdivision within Yobe State used in this study to represent origin or destination points for within-state transport routes.

Motor Park: A designated terminal or gathering point where buses load passengers, sell tickets, and coordinate departures.

Payment Gateway: An online service that processes electronic payments from passengers and confirms transaction status to the booking system.

QR Code: Quick Response code; a machine-readable barcode embedded in an electronic ticket to enable fast verification of booking authenticity at boarding points.

Real-Time Synchronization: Immediate updating of data across system components so that seat availability and booking status reflect current information without manual refresh delays.

Route: A defined transport path between an origin and a destination, with associated distance, fare, and operational rules.

Schedule: A planned departure and arrival timetable that links a specific bus to a route on a given date and time.

Seat Map: A visual layout of available and occupied seats in a vehicle, allowing passengers to select preferred seats during booking.

Transport Management System (TMS): An integrated software platform used to plan, coordinate, monitor, and control transport operations, including bookings, fleet activities, and reporting.

Vehicle Type: The category of transport vehicle assigned to a trip, such as Toyota bus, Sienna, Sharon, or Golf, each with distinct seating capacity and route suitability.

Web Application: A software system accessed through a web browser that provides interactive services to users over the internet without requiring local installation on the user device.

CHAPTER TWO

LITERATURE REVIEW

This chapter presents a review of relevant literature and theoretical concepts related to the design and implementation of an E-Transport Booking and Management System for Yobe State. The review examines scholarly and professional perspectives on the key variables embedded in the research title, namely e-transport, online booking, transport management systems, web-based information systems, and road transport operations. The purpose is to establish a conceptual foundation that guides the analysis, design, and implementation phases of this study (Dennis, Wixom, & Roth, 2014; Laudon & Laudon, 2020).

2.1 Conceptual Framework

The conceptual framework of this study identifies and defines the major concepts derived from the research topic. These concepts represent the core variables that the proposed system seeks to address. Each concept is discussed below in relation to its meaning, relevance, and application within the context of Yobe State transport services.

2.1.1 E-Transport

E-transport refers to the application of electronic and digital technologies to facilitate transport-related activities such as trip planning, reservation, payment, ticketing, fleet coordination, and service monitoring. It represents a shift from manual, paper-based transport processes to computerized platforms that improve speed, accuracy, and accessibility. E-transport is widely regarded as a component of e-commerce and e-government, where information systems are used to deliver public and commercial services through digital channels.

Conceptually, e-transport integrates hardware, software, communication networks, and data management tools to connect passengers, operators, and administrators. For passengers, it enables remote access to transport services; for operators, it supports efficient handling of bookings and resources; and for regulators, it provides structured data for oversight and planning. In developing economies, e-transport is particularly significant because it can extend service reach to users in both urban and semi-urban areas, provided there is reasonable internet connectivity and digital literacy.

In this study, e-transport is conceptualized as the overarching paradigm within which the proposed Yobe State platform operates. The system is not limited to ticket sales alone but encompasses the digital transformation of booking and management workflows across the transport value chain. This conceptualization aligns the project with contemporary efforts to modernize road transport services using web technologies and cloud-based infrastructure (World Bank, 2020; Federal Republic of Nigeria, 2019).

2.1.2 Online Booking System

An online booking system is an information system that allows users to search for available services, select options, make reservations, and confirm transactions through the internet. In the transport sector, online booking systems enable passengers to reserve seats on scheduled trips without physically visiting a terminal. Such systems typically provide interfaces for route selection, date and time choice, seat assignment, passenger information capture, and payment confirmation.

Conceptually, a booking system functions as a reservation control mechanism that links customer demand with available transport capacity. Its core elements include a service catalogue (routes and schedules), inventory management (seats or capacity), transaction processing (booking and payment), and confirmation output (ticket or receipt). Effective booking systems are designed to prevent duplicate reservations, maintain data integrity, and provide immediate feedback to users on booking status.

Within this research, the booking component is a primary variable because it directly addresses passenger-facing problems identified in Chapter One, such as long queues, uncertainty of seat availability, and inconvenience of manual ticketing. The conceptual framework therefore treats online booking as the front-end service layer of the e-transport platform, through which users interact with transport schedules and complete travel reservations in real time.

2.1.3 Transport Management System

A Transport Management System (TMS) is an integrated software solution used to plan, execute, and monitor transport operations. It supports administrative activities such as fleet registration, route definition, schedule creation, driver assignment, fare

management, booking oversight, and performance reporting. Unlike a standalone booking portal, a TMS provides backend control functions that enable transport organizations to manage resources, enforce operational policies, and analyze business performance.

Conceptually, TMS platforms serve as decision-support and operations-control systems. They convert fragmented operational data into structured information that managers can use to allocate buses efficiently, identify high-demand routes, track revenue, and reduce operational bottlenecks. Modern transport management systems also incorporate role-based access control, audit trails, and dashboard analytics to strengthen accountability and strategic planning.

In the context of this study, transport management is conceptualized as the administrative backbone of the proposed solution. While passengers interact with the booking interface, administrators use management modules to maintain master data, monitor bookings, verify tickets, and generate reports. This dual structure reflects the complete scope of the research title, which combines booking services with management capabilities in a single unified platform for Yobe State operators.

2.1.4 Web-Based Information Systems

Web-based information systems are software applications delivered through web browsers and supported by server-side processing, databases, and network infrastructure. They are characterized by platform independence, remote accessibility, ease of deployment, and support for multi-user interaction. In system development, the web paradigm has become a preferred approach for building scalable business and government applications because it reduces client-side installation requirements and simplifies maintenance.

For this project, the web-based approach is central because the target users, passengers and administrators, require accessible interfaces across desktop and mobile devices. The conceptual framework therefore includes system design and implementation as a technical variable that connects user requirements with the final deployed solution. Technologies such as Next.js, TypeScript, Supabase, and cloud hosting are applied within this framework to achieve a maintainable, secure, and extensible e-transport platform (Fielding, 2000; Vercel Inc., 2024; Supabase Inc., 2024).

2.1.5 Road Transport Operations in Yobe State

Road transport operations refer to the coordinated activities involved in moving passengers by road from one location to another using buses and other commercial vehicles. These operations include route planning, vehicle deployment, ticketing, boarding, departure control, and post-trip record keeping. In state-level contexts such as Yobe, road transport operations are influenced by geography, road infrastructure, vehicle types, fare policies, motor park organization, and seasonal travel patterns.

Conceptually, road transport operations in Yobe State comprise both interstate and intrastate services. Interstate services connect Yobe to major destinations including Abuja, Kano, Kaduna, Jos, Bauchi, Sokoto, Zamfara, Nasarawa, and Niger, while intrastate services link local government areas within the state. Operators often deploy different vehicle categories such as Toyota buses, Sienna, Sharon, and Golf based on route distance, road condition, and passenger volume. Some routes also impose vehicle restrictions to ensure safety and regulatory compliance.

2.2 Related Works and Research Gaps

This section reviews existing studies, commercial platforms, and technology-driven initiatives related to online transport booking and transport management systems. The review highlights what has been achieved in prior research and industry practice, and identifies gaps that justify the present study on a Yobe State E-Transport Booking and Management System.

Several researchers have investigated the application of information technology to public and commercial transport services. Studies in intelligent transport systems emphasize the use of computerized scheduling, reservation platforms, electronic fare collection, and data analytics to improve mobility and operational efficiency. In the context of developing countries, scholars have noted that digital transport solutions can reduce congestion at terminals, improve revenue accountability, and enhance passenger convenience when properly adapted to local infrastructure and user behaviour.

At the commercial level, a number of Nigerian transport companies have introduced online booking channels for major intercity routes. Companies such as God is Good

Motors (GIGM), ABC Transport, GUO Transport, and Peace Mass Transit provide web and mobile interfaces through which passengers can search trips, select seats, pay electronically, and receive booking confirmations for selected corridors. These platforms demonstrate the feasibility of e-ticketing in the Nigerian road transport sector and confirm growing passenger acceptance of digital booking services, particularly along high-traffic routes connecting Lagos, Abuja, Port Harcourt, and other southern and central urban centres.

Beyond individual operator platforms, third-party aggregators and travel marketplaces have emerged to compare routes, fares, and operators across regions. Such systems function as intermediaries between passengers and multiple transport providers, offering convenience and broader route visibility. Internationally, models such as intercity bus reservation portals, rail e-ticketing systems, and multimodal journey planners have shown that integrated digital booking significantly improves user experience and operator coordination when supported by reliable payment systems and centralized inventory control.

Research on transport management systems also shows that successful implementations combine passenger-facing services with backend operational modules. Systems that support only online ticket sales without administrative controls for fleet, route, and schedule management often fail to meet the full needs of transport operators. Conversely, management-heavy systems without intuitive passenger interfaces may not improve public service delivery. The literature therefore supports an integrated approach in which booking, ticketing, fleet administration, and reporting are implemented within a unified platform supported by a robust database and secure access control.

Research Gaps

Despite the progress recorded in related works and commercial deployments, several research and practice gaps remain relevant to the present study:

1. Most existing online booking solutions in Nigeria are operator-specific or regionally concentrated, with limited coverage of northeastern states such as Yobe. Many platforms are optimized for southern and central transport corridors

and do not reflect the route structure, fare patterns, and vehicle policies common in Yobe State.

2. There is a lack of localized systems that integrate both interstate and intrastate transport operations within a single platform tailored to Yobe local government area connections and state-level route management.
3. Few available platforms incorporate state-specific operational rules such as vehicle-type restrictions on designated long-distance routes, for example limiting certain destinations to approved bus categories.
4. Existing generic transport applications often provide booking interfaces without comprehensive administrative modules for managing companies, buses, drivers, schedules, ticket verification, and operational reports required by small and medium transport operators.
5. Manual booking and paper-based ticketing remain widespread in many Yobe motor parks, indicating a digital adoption gap that has not been adequately addressed by national or regional platforms designed for other markets.
6. Limited research has documented the design and implementation of web-based e-transport systems that apply modern cloud-native technologies such as Next.js, TypeScript, and Supabase to solve booking and management problems in a northeastern Nigerian state context.
7. There is insufficient emphasis in prior works on real-time seat synchronization, QR-based e-ticket verification, and combined passenger-admin workflows within one cohesive system designed for Yobe State transport stakeholders.

These gaps establish the need for the current research. The proposed Yobe State E-Transport Booking and Management System seeks to fill them by delivering a context-specific, integrated, and web-based solution that supports both passenger booking and administrative transport management. In doing so, the study extends existing knowledge and practice beyond generic platforms toward a practical digital infrastructure aligned with the operational realities of Yobe State.

2.3 Summary of the Review

This chapter has reviewed literature and conceptual foundations relevant to the design and implementation of a Yobe State E-Transport Booking and Management System. The review began with the conceptual framework, in which the major variables of the research

title were identified and defined. These variables include e-transport, online booking systems, transport management systems, web-based information systems, and road transport operations in Yobe State. Each concept was explained in relation to its theoretical meaning and practical significance for the proposed solution.

The review showed that e-transport represents the broader digital transformation of transport services through electronic channels, while online booking systems provide the passenger-facing mechanism for route search, seat reservation, payment, and ticket confirmation. Transport management systems complement booking services by offering administrative control over fleet, routes, schedules, drivers, bookings, and reporting. Web-based information systems provide the technical foundation for implementing such services in an accessible and scalable manner, and Yobe State road transport operations supply the real-world domain in which the system is applied.

In summary, the literature supports the need for an integrated, web-based e-transport platform tailored to Yobe State. The present study therefore builds on established concepts and prior efforts while addressing identified gaps through the design and implementation of a dedicated booking and management system using modern web technologies. The summary of the reviewed themes, findings, and gaps is presented in Table 2.1 below.

Table 2.1: Summary of Literature Review

S/N	Theme/Area Reviewed	Summary of Existing Knowledge	Relevance to the Study	Research Gap Identified
1	E-Transport	Digital technologies are widely used to automate transport booking, payment, ticketing, and	E-transport improves convenience, transparency, and data availability for operators and passengers.	Many implementations are generic and not adapted to Yobe State transport practices.

		operational coordination.		
2	Online Booking System	Web and mobile platforms enable remote seat reservation, fare payment, and ticket issuance.	Online booking reduces queues and improves passenger access to schedule information.	Existing booking platforms rarely cover Yobe routes and local motor park workflows comprehensively.
3	Transport Management System (TMS)	Integrated TMS solutions support fleet, route, schedule, driver, and booking administration.	Backend management improves accountability, reporting, and operational control.	Several public booking apps lack full administrative modules required by local operators.
4	Web-Based Information Systems	Modern web frameworks support scalable, cross-device access and cloud deployment.	Web systems are suitable for passengers and administrators using desktop or mobile browsers.	Limited studies apply current cloud-native stacks to northeastern Nigerian transport problems.
5	Nigerian Commercial Platforms	Operators such as GIGM, ABC Transport, GUO Transport, and Peace Mass Transit offer digital booking on major routes.	Commercial success confirms passenger acceptance of e-ticketing in Nigeria.	Coverage is concentrated on high-traffic corridors outside Yobe State.
6	Real-Time Seat Management	Concurrent booking environments	Real-time synchronization improves booking	Localized real-time seat maps for Yobe vehicle

		require live seat updates to prevent duplicate reservations.	accuracy and customer trust.	types are not widely available.
7	Electronic Ticketing and QR Verification	Digital tickets with QR codes enhance boarding verification and reduce ticket fraud.	E-tickets support paperless and verifiable travel documentation.	QR-enabled ticketing is not consistently integrated with local operator management systems in Yobe.
8	Yobe Road Transport Operations	Yobe transport includes interstate and intrastate movement using buses, Sienna, Sharon, and Golf vehicles.	Operational context defines routes, fares, vehicle rules, and administrative needs.	No dedicated e-transport system currently addresses this state-specific operational context.

CHAPTER THREE

SYSTEM ANALYSIS AND DESIGN

This chapter presents the system analysis and design of the Yobe State E-Transport Booking and Management System. It describes the methodology adopted for the study, the approach used to gather and analyse requirements, and the design framework applied before implementation. The chapter provides the foundation for the system development activities discussed in subsequent chapters.

3.1 Methodology Adopted

The methodology adopted for this study is the Object-Oriented System Development Life Cycle (OO-SDLC) combined with iterative prototyping (Sommerville, 2016; International Organization for Standardization, 2018). This approach was selected because software projects of this nature require structured progression from problem identification to deployment, while also allowing incremental refinement of system components based on testing and supervisor feedback. The OO-SDLC provides a disciplined framework for analysing user requirements, modelling system behaviour, designing databases and interfaces, implementing modules, and validating the final product.

Object-oriented system development was considered appropriate for this project because the e-transport platform comprises distinct but interacting entities such as passengers, administrators, buses, routes, schedules, seats, bookings, payments, and tickets. Modelling the system using objects, classes, relationships, and use cases makes it easier to represent real-world transport operations and translate them into software components. In addition, the web application architecture adopted in this study naturally aligns with object-oriented and component-based design principles.

The iterative prototyping dimension of the methodology enabled the researcher to develop core modules in stages, test them early, and improve functionality before full integration. Rather than waiting until the end of development to evaluate the system, prototypes of key features such as route search, seat map selection, booking workflow, admin dashboard, and ticket generation were built, reviewed, and refined progressively.

This reduced development risk and ensured that the final system remained aligned with the operational needs of Yobe State transport services.

The methodology was implemented through the following phases:

1. Problem identification and requirements gathering: Existing manual transport booking and management practices in Yobe State were examined, and functional as well as non-functional requirements were identified for passengers and administrators.
2. System analysis: User roles, system processes, data flows, and constraints were analysed to determine what the system must do and the conditions under which it must operate.
3. System design: UML-based models including use case, activity, sequence, and entity-relationship diagrams were prepared to represent system structure, interaction, and database relationships (Object Management Group, 2017; Gamma et al., 1994).
4. Database design: A relational database schema was designed to store companies, buses, routes, schedules, seats, bookings, payments, users, and related records (Connolly & Begg, 2015; PostgreSQL Global Development Group, 2024).
5. Implementation: The system was developed as a web application using Next.js, TypeScript, Supabase, and supporting libraries for maps, payments, email, and ticket generation.
6. Testing and evaluation: Functional, usability, and integration tests were conducted to verify that system modules performed according to specification.
7. Documentation and deployment preparation: The implemented system was documented and prepared for deployment on a cloud hosting environment.

The following tools and techniques supported the adopted methodology:

- Visual Studio Code as the integrated development environment (IDE) for coding, debugging, and project management.
- Next.js and TypeScript for full-stack web application development.
- Supabase (PostgreSQL) for database creation, authentication, and backend data services.

- Draw.io or similar CASE tools for preparing UML and entity-relationship diagrams during the design phase.
- Git and GitHub for version control and collaborative tracking of development changes.
- Browser-based testing and manual test scenarios for validating booking, payment, ticketing, and administrative workflows.

In summary, the OO-SDLC with iterative prototyping provided a suitable methodology for this study because it combined formal system analysis and design with practical incremental development. This ensured that the resulting Yobe State E-Transport Booking and Management System was not only technically sound but also responsive to the real operational requirements identified during the research process.

3.2 Analysis of the Existing System

The analysis of the existing system examines how transport booking and management activities are currently carried out in Yobe State motor parks and booking offices before the introduction of the proposed digital platform. This analysis is essential because it reveals the processes, actors, data handling methods, and operational weaknesses that the new system is intended to address. The existing system referred to in this study is the conventional manual transport management approach used by many road transport operators in the state.

In the current arrangement, passengers who wish to travel must physically visit a motor park or booking counter to make enquiries about available trips. A booking clerk or agent receives the passenger request, checks seat availability using a paper manifest, notebook, or verbal confirmation from colleagues, and communicates the fare for the selected route. If a seat is available, the passenger pays in cash and receives a handwritten or pre-printed paper ticket. The passenger name and seat number are then entered manually into a manifest register. This process is repeated for every traveller and for every booking point within the terminal.

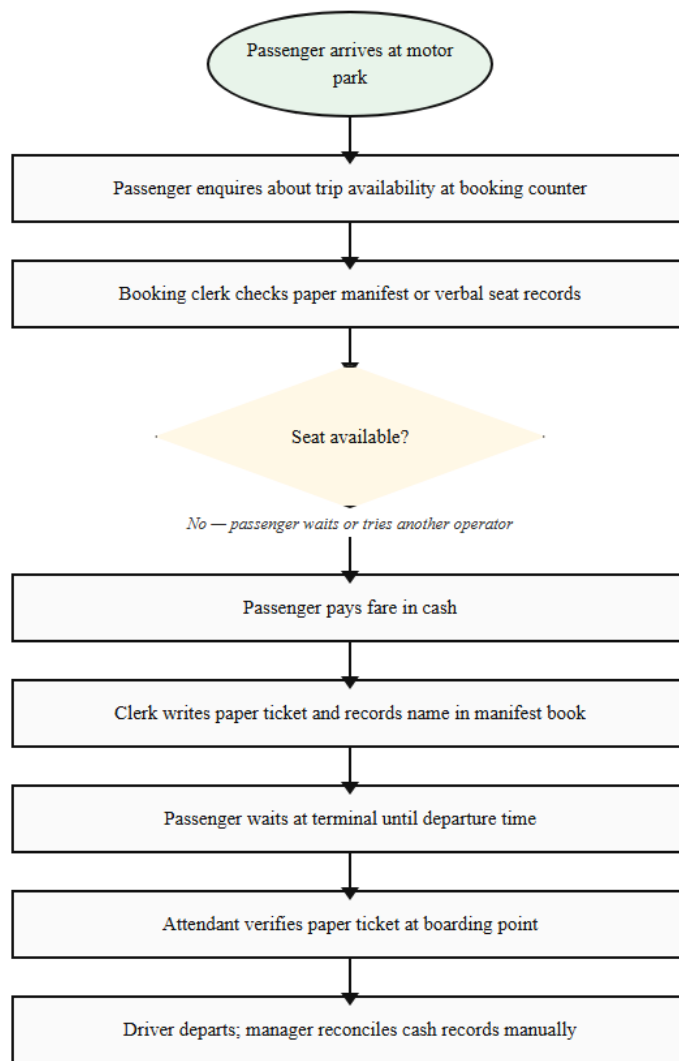


Figure 3.1: Existing Manual Transport Booking Workflow

The existing system involves several stakeholders. The passenger initiates the travel request. The booking clerk or agent handles enquiries, fare collection, and ticket issuance. Motor park attendants coordinate vehicle loading and boarding. Drivers and conductors manage departure and on-road passenger control. Transport managers supervise daily operations and review cash collections and trip records at the close of business. All these actors depend on paper records such as manifest books, ticket stubs, fare sheets, and cash ledgers. The workflow of the existing manual booking process is illustrated in Figure 3.1.

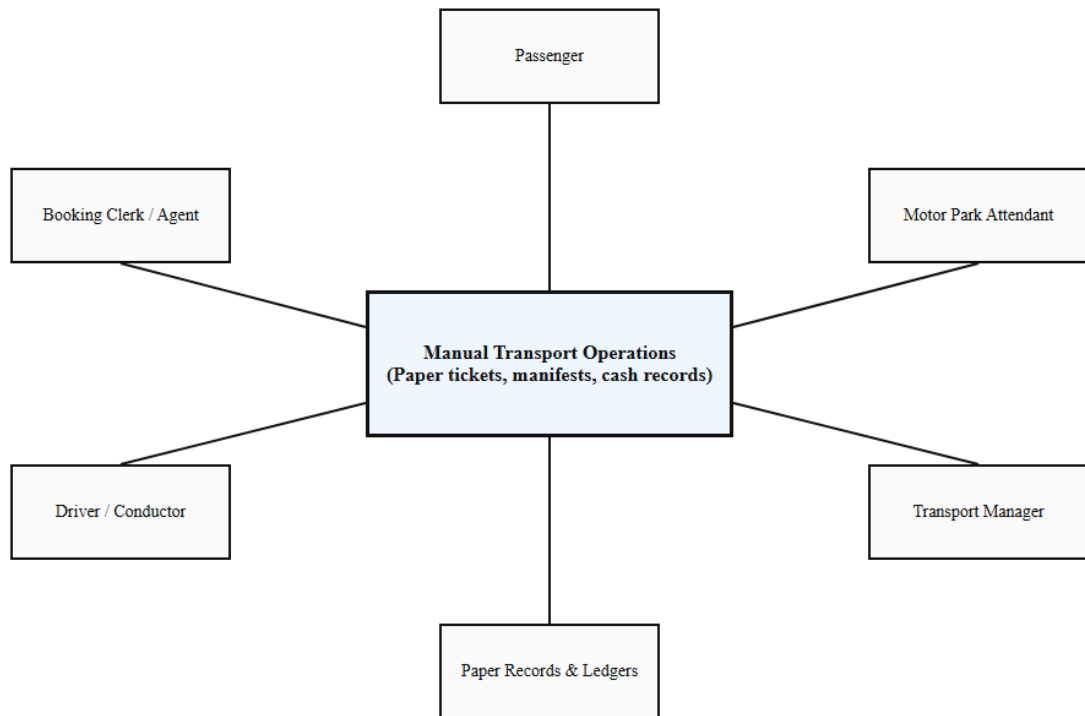


Figure 3.2: Stakeholders in the Existing Manual Transport System

Figure 3.1 shows that the manual system is sequential and labour-intensive. Each stage depends on human intervention, physical presence at the terminal, and manual verification of information. There is no automated synchronization of seat availability across counters, and passengers who are not physically present cannot complete bookings in advance. The stakeholder structure of the existing system is presented in Figure 3.2, which highlights the central role of manual coordination and paper documentation.

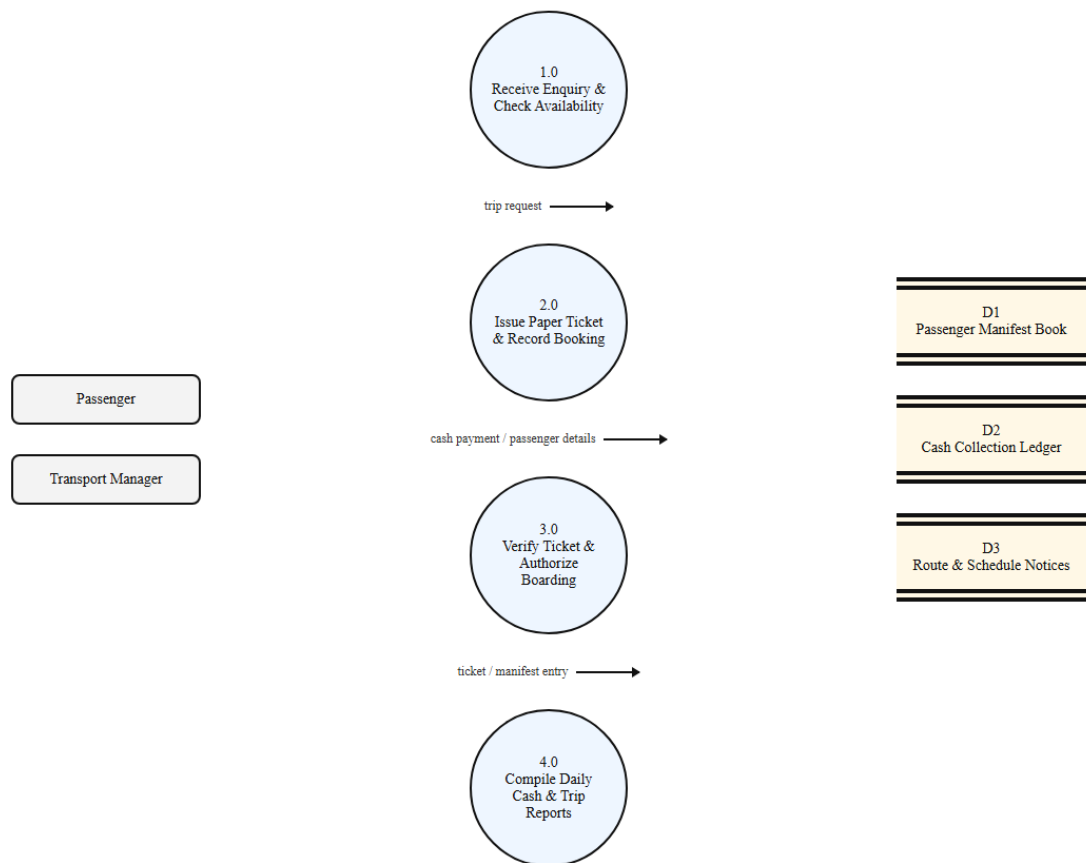


Figure 3.3: Data Flow in the Existing Manual Transport System

From a data management perspective, the existing system stores information in fragmented physical formats. Passenger manifest books serve as the main record of bookings, while cash ledgers track daily collections. Route and schedule information may be displayed on notice boards or communicated verbally to booking staff. Figure 3.3 presents a simplified Level guides the analysis, design, and implementation phases of this study (Dennis, Wixom, & Roth, 2014; Laudon & Laudon, 2020). data flow representation of the manual system, showing how information moves between passengers, clerks, attendants, managers, and paper-based data stores.

The analysis identified several weaknesses in the existing system. First, information processing is slow because every booking requires manual checking and writing. Second, seat allocation is prone to error, especially when multiple clerks serve passengers simultaneously during peak periods. Third, ticket verification depends on visual inspection of paper documents, which can be lost, duplicated, or falsified. Fourth, managers lack real-time access to booking statistics, route occupancy, and revenue

summaries. Fifth, passengers have limited visibility into available trips, vehicle types, and departure schedules unless they visit the terminal in person.

Another limitation of the existing system is poor integration between booking, payment, fleet management, and reporting functions. Each activity is handled as a separate manual task, making it difficult to generate accurate operational reports or enforce route-specific vehicle policies. For example, restrictions on certain vehicle types for long-distance routes must be remembered and applied manually by staff. Likewise, reconciliation of daily cash receipts with manifest entries is time-consuming and vulnerable to omission or miscalculation.

Despite these weaknesses, the existing system has some strengths that should be acknowledged. It requires minimal initial technology investment, is easy for staff with low digital literacy to understand, and supports direct personal interaction between passengers and booking agents. Cash payments are widely accepted in the local transport environment, and paper tickets are familiar to many travellers. However, these advantages are increasingly outweighed by the demand for faster service, better record keeping, and improved passenger convenience.

In conclusion, the analysis of the existing system confirms that the current manual approach is no longer adequate for efficient transport service delivery in Yobe State. The process is characterized by fragmented data handling, heavy dependence on paper records, limited administrative control, and poor support for remote or advance booking. These findings provide a clear basis for designing the proposed E-Transport Booking and Management System, which seeks to automate booking workflows, centralize records, improve seat management, and provide digital tools for both passengers and administrators.

3.3 The Proposed System Design

The proposed system is a web-based E-Transport Booking and Management System designed to replace the manual processes identified in Section 3.2 with an integrated digital platform for Yobe State road transport operations. The system provides separate but connected interfaces for passengers and administrators, supported by a centralized database and external service integrations. Its design emphasizes usability, security,

scalability, and alignment with the operational realities of interstate and intrastate transport in Yobe State.

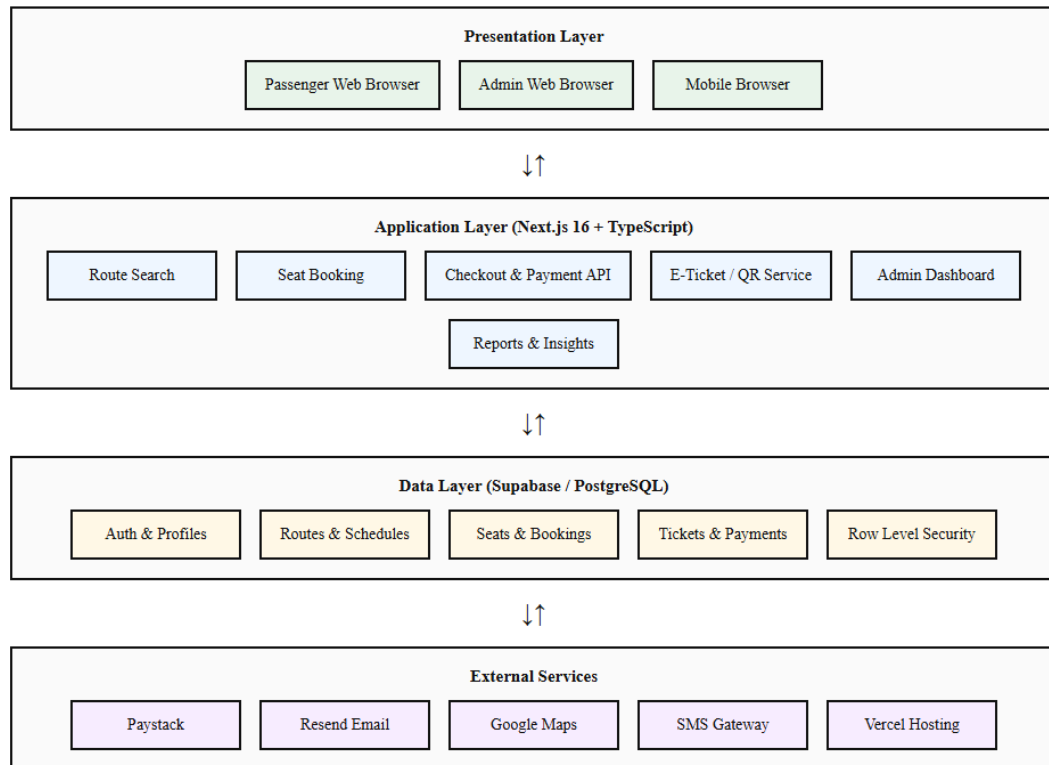


Figure 3.4: Proposed System Architecture

The proposed solution adopts a three-tier architecture consisting of a presentation layer, an application layer, and a data layer, with optional external services for payments, notifications, and mapping. The presentation layer is accessed through standard web browsers on desktop and mobile devices. The application layer is implemented using Next.js 16 with TypeScript and handles route search, seat booking, checkout, ticket generation, and administrative operations. The data layer is powered by Supabase, which provides PostgreSQL database storage, authentication, and row-level security. The overall architecture of the proposed system is shown in Figure 3.4.

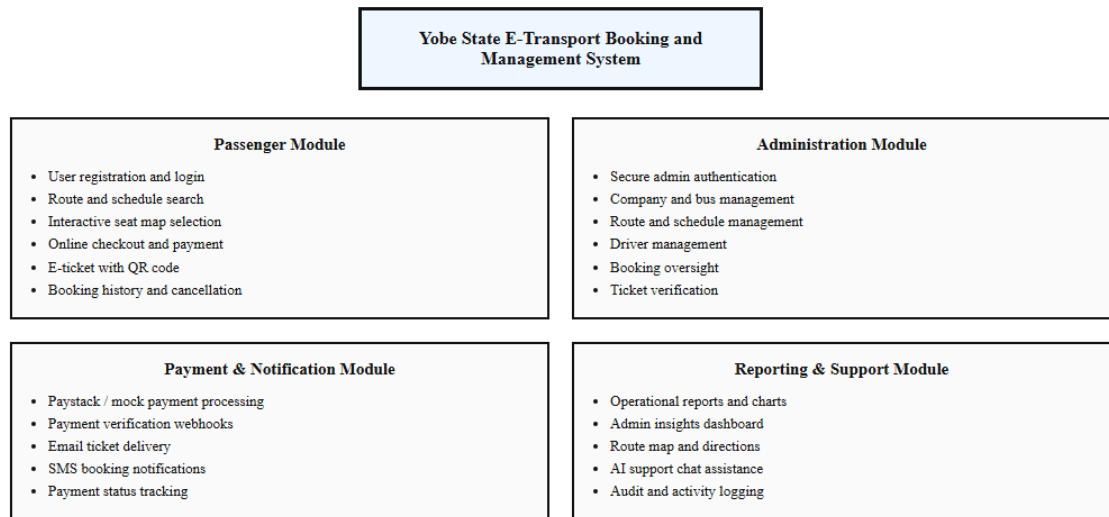


Figure 3.5: Functional Modules of the Proposed System

Functionally, the proposed system is organized into modular components to improve maintainability and clarity of responsibilities. The passenger module supports account registration, trip search, seat selection, payment, and access to electronic tickets. The administration module provides secure login, management of companies, buses, drivers, routes, schedules, and bookings, as well as ticket verification at boarding points. The payment and notification module handles online payment processing, email and SMS ticket delivery, and transaction status tracking. The reporting and support module provides charts, operational summaries, route maps, and optional AI-assisted support. The functional decomposition of the proposed system is illustrated in Figure 3.5.

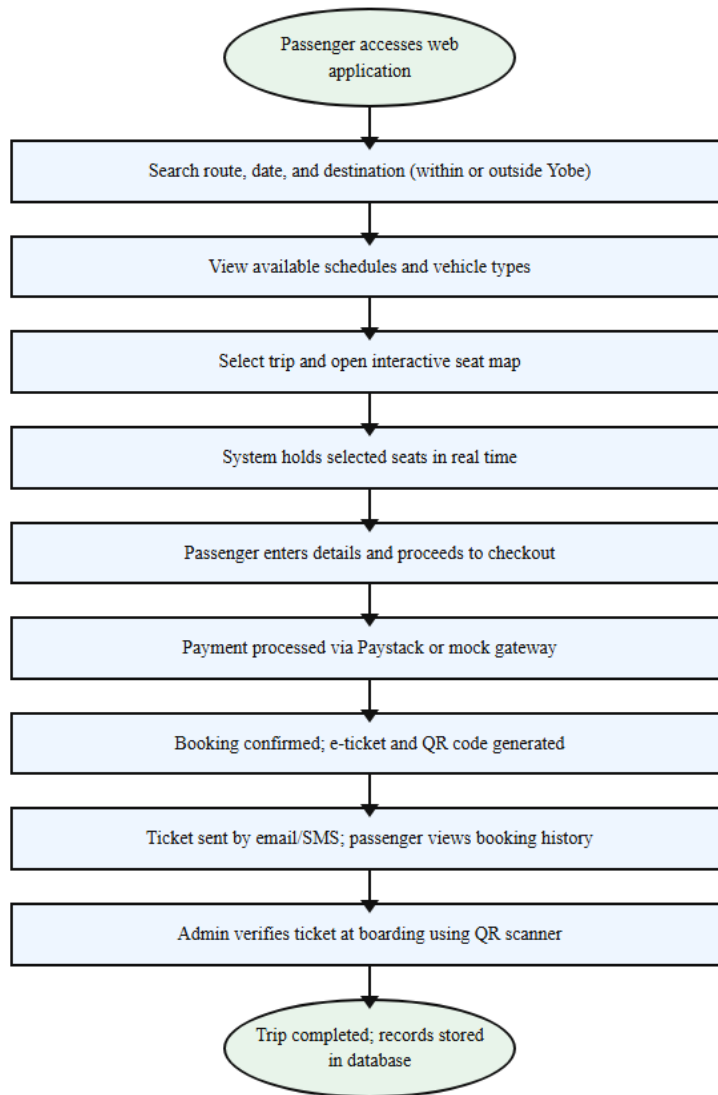


Figure 3.6: Proposed Online Booking Workflow

The proposed booking workflow is designed to automate the full passenger journey from trip search to boarding verification. A passenger first accesses the application and searches for routes by origin, destination, and travel date. The system returns available schedules filtered by vehicle type and route rules applicable to Yobe State operations. The passenger selects a trip, views a top-view seat map, and chooses available seats. The system temporarily holds the selected seats to prevent duplicate booking while the passenger completes checkout. After successful payment, the booking is confirmed, an electronic ticket with QR code is generated, and notifications are sent to the passenger. Administrators can verify tickets digitally before boarding. The proposed online booking workflow is presented in Figure 3.6.

The database design of the proposed system is relational and normalized around core transport entities. Major tables include companies, profiles, buses, routes, schedules, seats, bookings, booking_seats, tickets, and payments. User roles are defined as passenger, staff, and admin, with access control enforced through authentication and database policies. Seat status is maintained as available, held, or booked, enabling real-time inventory management. Booking and payment statuses are tracked throughout the transaction lifecycle to support confirmation, cancellation, and reporting.

The proposed system also incorporates Yobe-specific business rules in its design. Routes originate from Yobe State and extend to destinations such as Abuja, Jos, Bauchi, Kano, Kaduna, Nasarawa, Niger, Sokoto, and Zamfara, as well as connections between local government areas within the state. Vehicle types including Toyota bus, Sienna, Sharon, and Golf are supported with capacity-based seat layouts. Route-specific restrictions, such as limiting certain long-distance trips to approved vehicle categories, are enforced at the application level during search and schedule display.

Security is an important part of the proposed design. Passenger and admin access is protected through authenticated sessions. Sensitive administrative functions are restricted to authorized users. Payment references are validated before booking confirmation. QR payloads on tickets are used to support tamper-resistant verification. The use of Supabase row-level security further ensures that users can only access records permitted by their role.

The user interface design of the proposed system follows a responsive web layout using Tailwind CSS. Passenger pages prioritize simple search forms, schedule cards, visual seat maps, and clear checkout steps. Admin pages use a dashboard layout with sidebar navigation, tabular data views, forms for resource management, and charts for reporting. This design approach ensures that the system remains usable on both large screens and mobile devices commonly used by passengers in the target environment.

In summary, the proposed system design provides a comprehensive, modular, and technically sound framework for digitizing transport booking and management in Yobe State. It addresses the weaknesses of the existing manual system by centralizing data,

automating workflows, supporting real-time seat management, and enabling digital ticketing and administrative oversight. The subsequent sections of this chapter and the implementation chapter that follows build upon this design foundation through detailed requirements specification, modelling diagrams, and system development.

3.4 The Architecture, Model, and Design Specifications

This section presents the architectural and design models used to specify the structure, behaviour, and data relationships of the proposed Yobe State E-Transport Booking and Management System. These models translate the high-level system design described in Section 3.3 into formal representations that guided database creation, application development, and system testing. The models include the component architecture, use case model, entity-relationship model, and sequence model for the booking process.

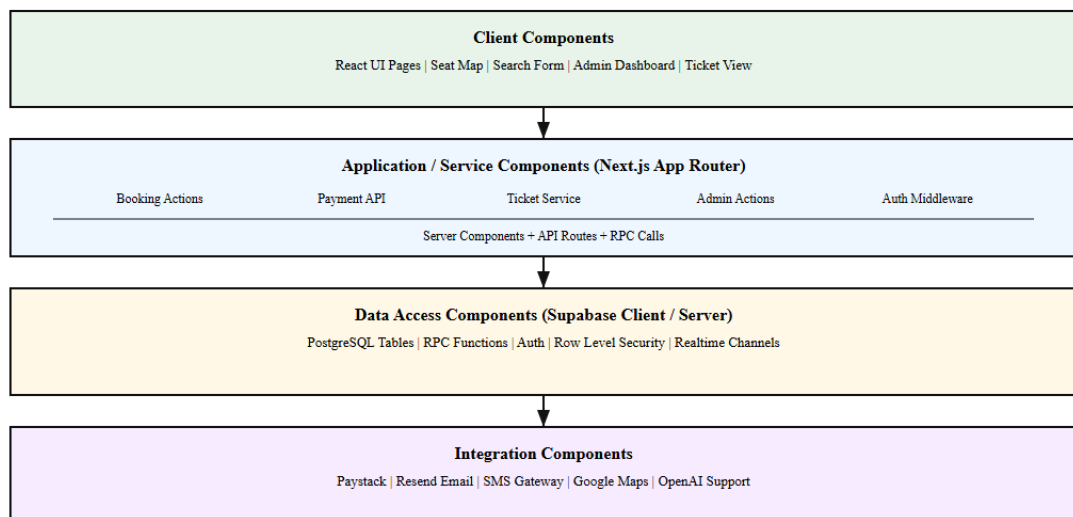


Figure 3.7: Component Architecture Model

The architectural model of the proposed system follows a layered client-server approach implemented on the web. At the highest level, the system operates as a browser-based client communicating with a Next.js application server, which in turn communicates with a Supabase backend and selected external services. This model separates concerns among user interface components, business logic, data persistence, and third-party integrations. Such separation improves maintainability, security, and scalability. The component architecture model is shown in Figure 3.7.

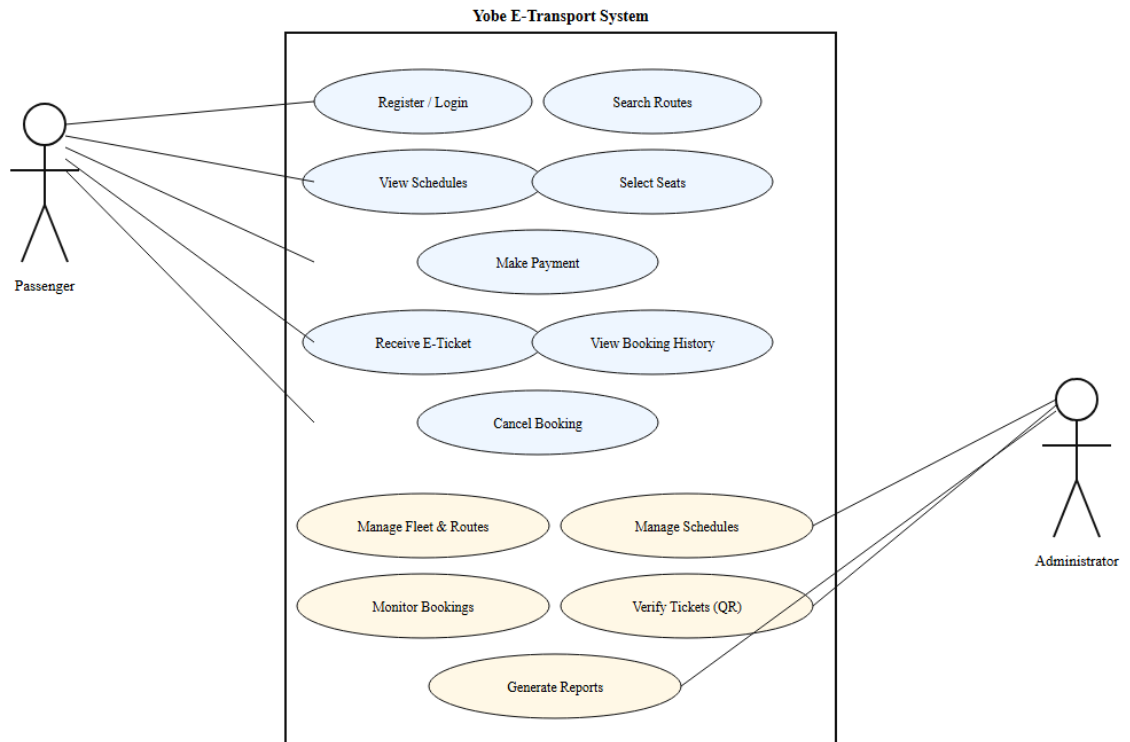


Figure 3.8: Use Case Diagram of the Proposed System

The component model identifies four major groups of system elements. Client components include React-based pages and visual widgets such as the search form, seat map, booking list, and admin dashboard. Application components include booking actions, payment APIs, ticket services, authentication middleware, and administrative operations implemented within the Next.js App Router. Data access components include Supabase client libraries, SQL tables, stored procedures, and row-level security policies. Integration components connect the application to Paystack for payments, Resend for email, SMS gateways, Google Maps for route visualization, and optional OpenAI support services.

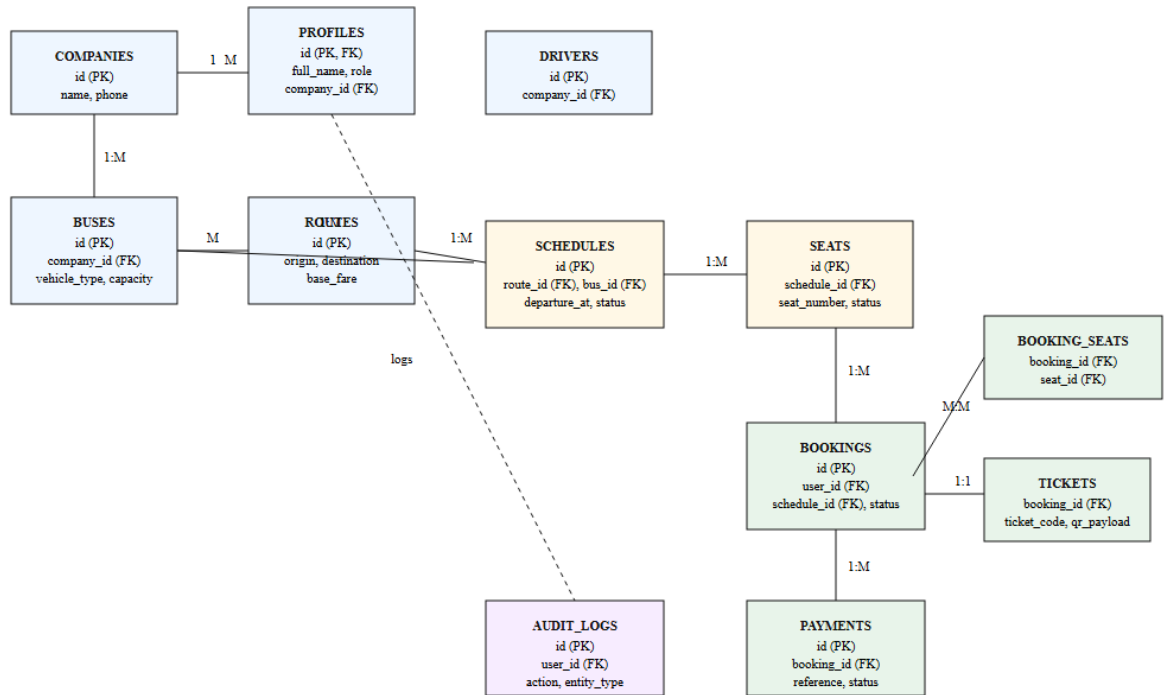


Figure 3.9: Entity-Relationship Diagram (Core Data Model)

The behavioural model of the system is represented using a UML use case diagram. Two primary actors interact with the system: the passenger and the administrator. Passenger use cases include registration, login, route search, schedule viewing, seat selection, payment, e-ticket receipt, booking history review, and booking cancellation. Administrator use cases include fleet and route management, schedule management, booking monitoring, QR ticket verification, and report generation. The use case diagram in Figure 3.8 defines the functional boundaries of the system and serves as the basis for requirements validation.

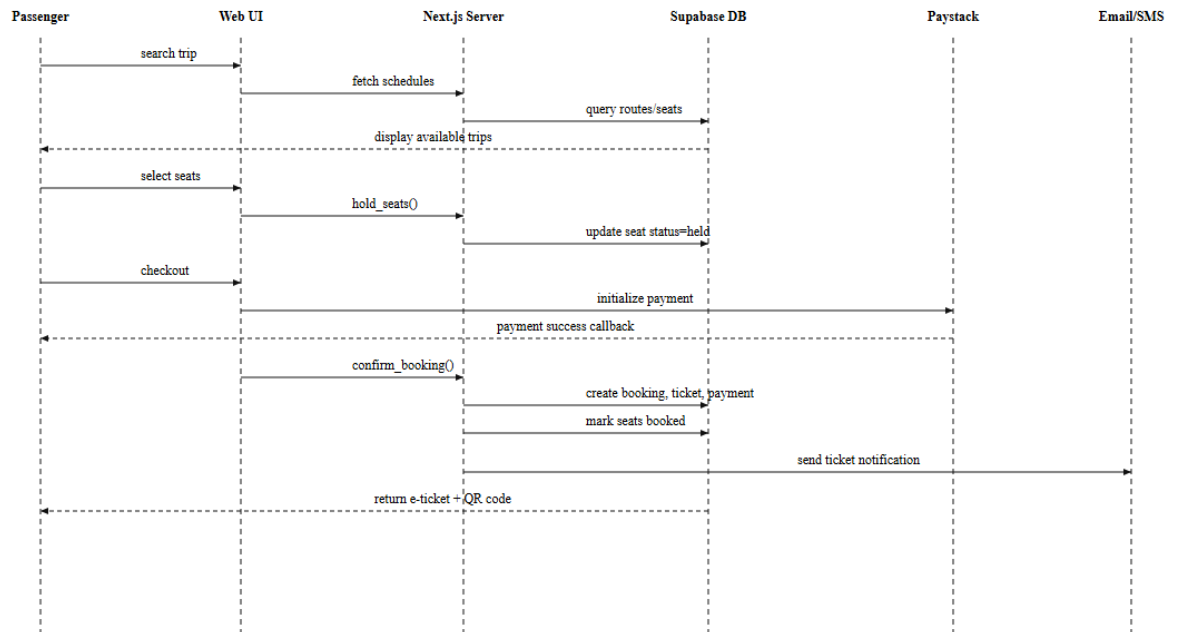


Figure 3.10 Sequence Diagram for Online Booking Process

The data model of the proposed system is represented using an entity-relationship diagram (ERD). The ERD captures the main entities and relationships required to support transport operations. A company owns multiple buses, routes, and drivers. A route and a bus are linked through a schedule, and each schedule generates multiple seats. A passenger profile creates bookings against schedules, and each booking may reserve one or more seats through the `booking_seats` associative entity. Successful bookings produce tickets and payment records. Audit logs capture significant administrative and system actions. The core ERD is presented in Figure 3.9.

The booking process model is represented using a UML sequence diagram to show runtime interaction among system objects during an online reservation. The passenger initiates a trip search through the web interface. The server queries the database for available schedules and seats. When seats are selected, the `hold_seats` stored procedure temporarily reserves them. During checkout, payment is processed through Paystack or a mock gateway. Upon successful payment, the `confirm_booking` procedure creates the booking record, marks seats as booked, generates a ticket with QR payload, and triggers email or SMS notification. The sequence diagram in Figure 3.1 guides the analysis, design, and implementation phases of this study (Dennis, Wixom, & Roth, 2014; Laudon & Laudon, 2020). illustrates this interaction flow.

In addition to these graphical models, the system uses a relational database schema implemented in PostgreSQL through Supabase migrations. Enumerated types define controlled values for user roles, seat status, booking status, schedule status, payment status, bus status, and vehicle type. Database triggers automatically generate seats when schedules are created and maintain updated timestamps on mutable records. Stored procedures such as `hold_seats`, `confirm_booking`, and `cancel_booking` encapsulate critical booking logic and help prevent race conditions during concurrent seat selection.

The deployment model of the proposed system targets cloud hosting on Vercel for the web application, with Supabase providing managed database and authentication services. This model supports remote access by passengers and administrators, simplified deployment from the development environment, and separation of frontend hosting from backend data services. Environment variables are used to configure secrets such as database keys, admin passwords, payment credentials, and notification service tokens.

Overall, the architecture and design models provide a structured blueprint for the proposed system. They define how user interfaces, business services, database entities, and external integrations cooperate to deliver a complete e-transport solution. These models also create a basis for implementation, testing, and future system extension in Yobe State transport operations.

3.5 Design Flowchart and Algorithm

Flowcharts and algorithms are essential design tools for representing the logical flow of operations within the proposed system. They provide step-by-step descriptions of how the application processes user requests, makes decisions, and updates system data. In this study, flowcharts were used to model the main booking process and the core database algorithms that support seat reservation and booking confirmation. Pseudocode representations are also provided to describe the logic implemented in stored procedures and server actions.

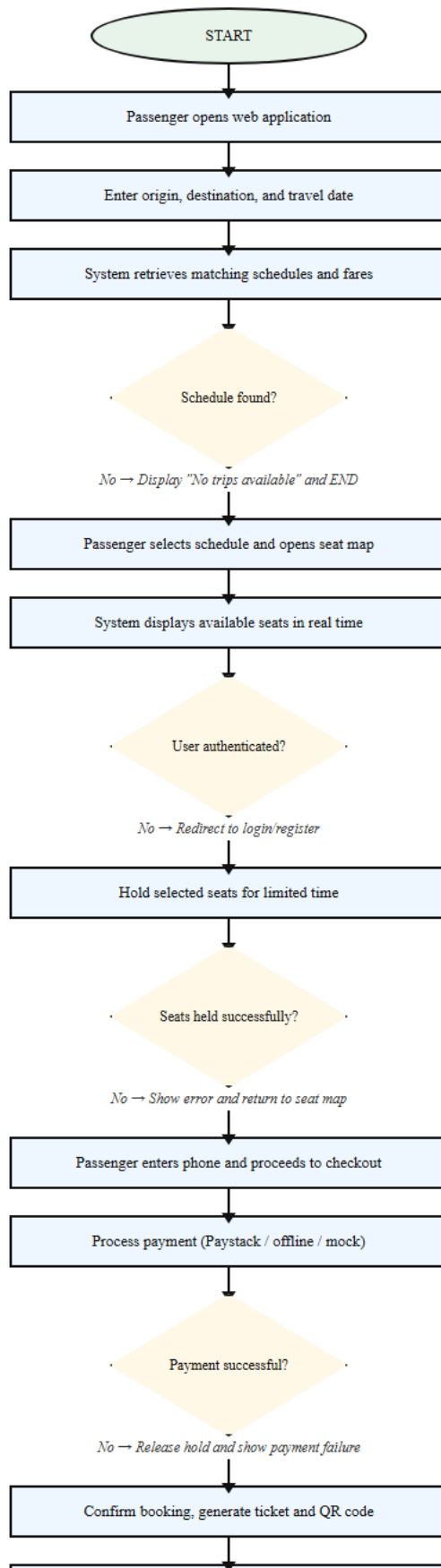


Figure 3.11: Main System Flowchart for Online Booking

The main system flowchart describes the end-to-end passenger booking process from application access to ticket issuance. The process begins when a passenger opens the web application and enters trip search criteria such as origin, destination, and travel date. The system retrieves matching schedules and displays available trips. The passenger selects a schedule, views the seat map, and chooses one or more seats. If the user is not authenticated, the system redirects to the login or registration page. After seat selection, the system temporarily holds the selected seats while the passenger completes checkout and payment. If payment is successful, the booking is confirmed, a ticket with QR code is generated, and notifications are sent to the passenger. The main system flowchart is presented in Figure 3.11.

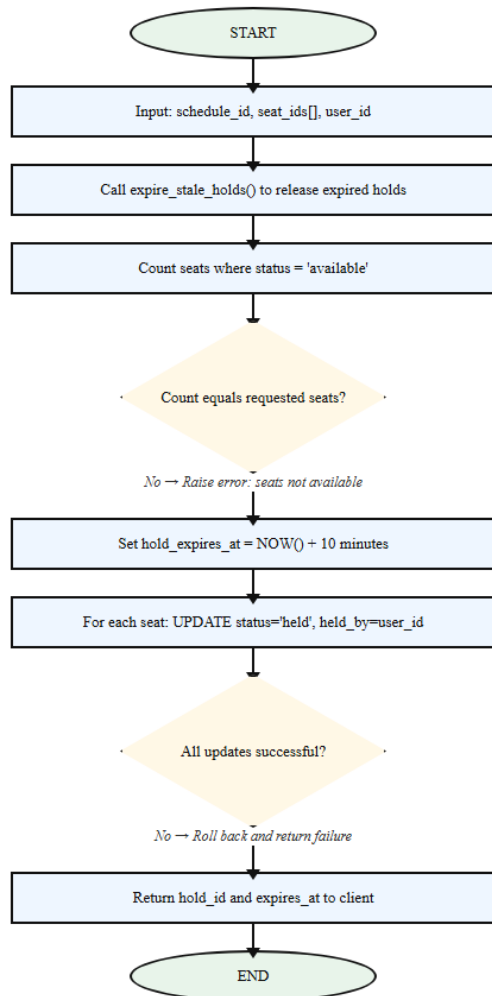


Figure 3.12: Flowchart for Seat Hold Algorithm

The seat hold algorithm is a critical part of the booking design because it prevents multiple passengers from selecting the same seat at the same time. When a passenger selects seats, the system calls the `hold_seats` stored procedure with the schedule identifier, seat identifiers, and user identifier. The algorithm first expires stale holds that have passed their time limit. It then verifies that all requested seats are still available. If validation succeeds, the seats are marked as held, associated with the current user, and assigned an expiry time of ten minutes. If any seat cannot be held, the operation fails and the user is prompted to choose different seats. The seat hold algorithm flowchart is shown in Figure 3.12.

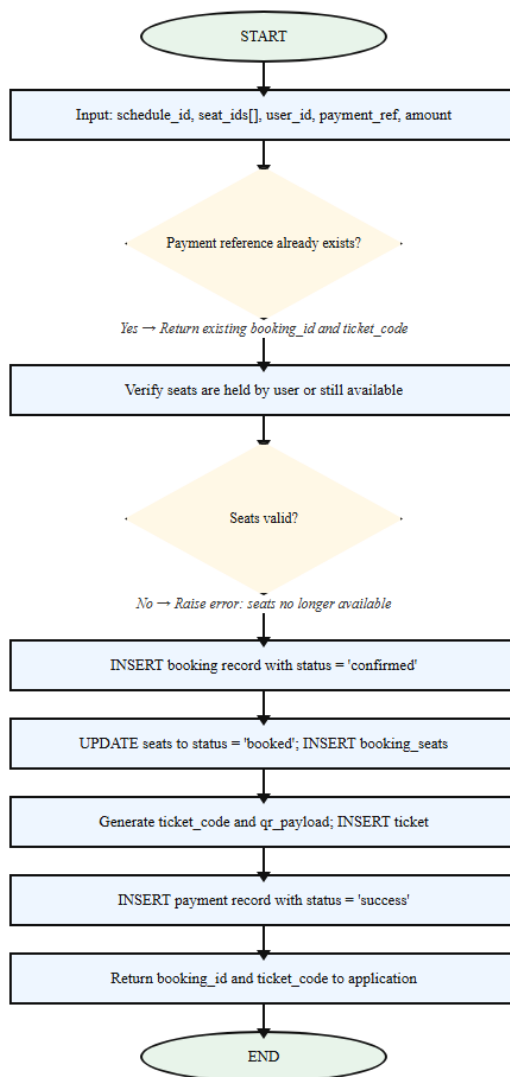


Figure 3.13: Flowchart for Booking Confirmation Algorithm

The booking confirmation algorithm executes after successful payment. It receives the schedule, seat list, user details, payment reference, amount, and payment provider. The algorithm first checks whether the payment reference already exists to ensure idempotent processing. It then verifies that the selected seats are either held by the current user or still available. A booking record is created with confirmed status, the seats are marked as booked, and entries are inserted into the booking_seats table. A unique ticket code and QR payload are generated and stored in the tickets table, while the payment record is saved with success status. The algorithm returns the booking identifier and ticket code to the application layer. The booking confirmation flowchart is illustrated in Figure 3.13.

The pseudocode below summarizes the main booking confirmation logic implemented in the system:

Algorithm: ConfirmBooking(schedule_id, seat_ids, user_id, payment_ref, amount)

BEGIN

IF payment_ref already exists THEN

RETURN existing booking_id and ticket_code

END IF

CALL ExpireStaleHolds()

IF seats are not held by user and not available THEN

RETURN error "Seats are no longer available"

END IF

CREATE booking with status = confirmed

FOR EACH seat_id IN seat_ids DO

UPDATE seat status = booked

INSERT booking_seat record

END FOR

GENERATE ticket_code and qr_payload

INSERT ticket and payment records

RETURN booking_id and ticket_code

END

In addition to booking-related algorithms, the system includes a cancellation algorithm that allows passengers or authorized administrators to cancel confirmed bookings. The

cancel_booking procedure verifies ownership or admin rights, releases booked seats back to available status, updates the booking status to cancelled, and marks related payments as refunded where applicable. This ensures that seat inventory remains accurate when bookings are reversed.

Overall, the flowcharts and algorithms presented in this section define the operational logic of the proposed system. They guided the implementation of stored procedures, server actions, and user interface workflows, and they provide a clear reference for testing and future maintenance of the Yobe State E-Transport Booking and Management System.

3.6 Advantages of the New System

The proposed Yobe State E-Transport Booking and Management System offers significant advantages over the existing manual transport operations described in Section 3.2. These advantages arise from the automation of booking workflows, centralization of operational data, and introduction of digital tools for passengers and administrators. The new system is designed not merely to replicate manual processes in electronic form, but to improve efficiency, accuracy, transparency, and service quality across the transport value chain.

One major advantage of the new system is improved passenger convenience. Travellers can search routes, view schedules, select seats, and complete bookings from any location with internet access. This eliminates the need for repeated physical visits to motor parks, reduces waiting time at booking counters, and allows passengers to plan trips in advance. Such convenience is particularly beneficial for residents in remote local government areas who would otherwise travel long distances to secure tickets in person.

Another important advantage is real-time seat management. In the manual system, seat availability depends on verbal confirmation or handwritten records that may not be synchronized across booking points. The proposed system uses a centralized database and seat hold mechanism to ensure that once a seat is selected or booked, it is immediately reflected across the platform. This reduces overbooking, seat conflicts, and passenger disputes during boarding.

The new system also improves ticketing security and verification. Instead of paper tickets that can be lost, damaged, or forged, passengers receive electronic tickets with unique codes and QR payloads. Administrators can verify tickets digitally at boarding points, improving accountability and reducing fraud. Passengers can also retain digital copies of their tickets and booking history for future reference.

From an administrative perspective, the proposed system provides centralized control over companies, buses, drivers, routes, schedules, and bookings. Managers can monitor operations through a dashboard, generate reports, and make data-driven decisions on fleet deployment and route performance. Manual reconciliation of cash and manifest records is replaced by structured digital transactions and automated summaries, thereby reducing administrative workload and error rates.

The system further supports better financial tracking and auditability. Payment records are linked directly to bookings, with references and statuses stored in the database. This improves revenue accountability and simplifies end-of-day reporting. Audit logs also capture important system actions, supporting transparency and internal control.

An additional advantage is the localization of the solution for Yobe State transport operations. The proposed system incorporates state-specific routes, vehicle types, fare structures, and route restrictions that generic national platforms may not support. This makes the system more relevant and practical for operators and passengers within the state.

The new system also enhances scalability and future expansion. Because it is web-based and cloud-hosted, additional routes, operators, and features can be introduced without rebuilding the entire platform. Integration with payment gateways, email, SMS, and mapping services provides a foundation for continuous improvement and modernization of transport services in Yobe State.

In summary, the proposed system delivers measurable improvements over the manual approach in terms of convenience, accuracy, security, administration, and long-term sustainability. A comparison of the existing manual system and the proposed system is presented in Table 3.2.

Table 3.2: Comparison of Existing Manual System and Proposed System

Aspect	Existing Manual System	Proposed E-Transport System
Booking access	Physical visit to motor park required	Online access from web browser anytime
Seat allocation	Manual and prone to duplication	Real-time centralized seat map with hold control
Ticketing	Paper tickets	Electronic tickets with QR verification
Payment tracking	Cash-based with manual reconciliation	Digital payment records linked to bookings
Record keeping	Manifest books and ledgers	Centralized database with audit logs
Reporting	Manual compilation	Automated dashboard and reports
Administrative control	Fragmented across staff	Unified admin module for fleet and operations
Passenger convenience	Low, especially for remote users	High, supports advance and remote booking
Data accuracy	Susceptible to human error	Improved through validation and automated workflows
Scalability	Limited by manual capacity	Expandable through cloud-based web architecture

CHAPTER FOUR

SYSTEM IMPLEMENTATION

This chapter describes the implementation of the Yobe State E-Transport Booking and Management System. It presents the user interfaces of the major system modules and provides summary code snippets that show how each module was developed. The implementation was carried out using Next.js 16, TypeScript, Tailwind CSS, and Supabase, with Visual Studio Code as the development environment.

The system was implemented in modular form to separate passenger services from administrative services while sharing a common database and authentication layer. The modules presented in this chapter include home and route search, trip results, seat selection, checkout and booking, e-ticket generation, admin login, and admin dashboard management.

4.1 Home and Route Search Module

The home and route search module provides the main entry point for passengers. It displays the application branding, trip scope selection (within Yobe or outside Yobe), origin and destination fields, travel date, and optional vehicle type filter. When the passenger submits the search form, the application navigates to the search results page with the selected parameters. This module was implemented using the SearchForm React component and Next.js client-side routing.

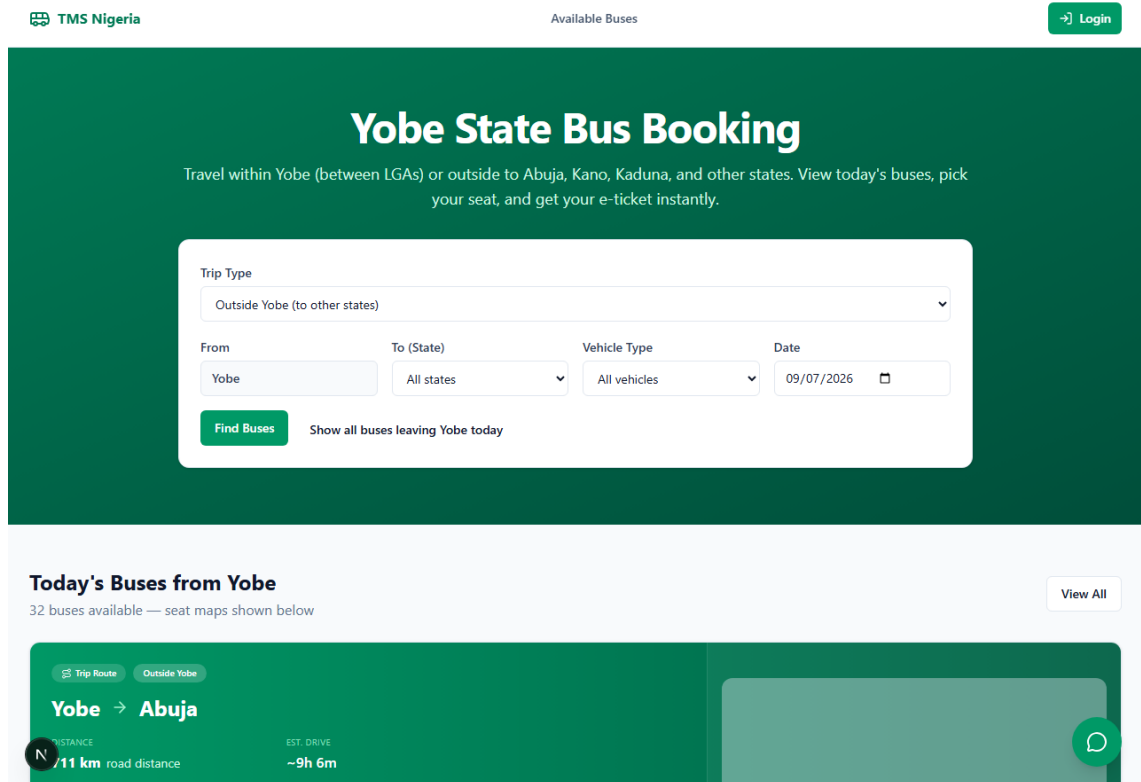


Figure 4.1: Home and Route Search Module Interface

Summary Code:

```
// src/components/search/search-form.tsx (summary)
function handleSubmit(e: React.FormEvent<HTMLFormElement>) {
  e.preventDefault();
  const form = new FormData(e.currentTarget);
  const date = (form.get("date") as string) || today;
  const params = new URLSearchParams({ scope, origin, destination, date });
  router.push(`/search?${params.toString()}`);
}
```

4.2 Trip Search Results Module

The search results module retrieves schedules that match the passenger query and displays them as trip cards showing route, departure time, fare, vehicle type, and available seats. Passengers can open a selected trip to proceed to seat selection. This module connects the search interface to the booking workflow and uses server-side queries against routes and schedules tables.

TMS Nigeria

Available Buses

Login

Available Buses

Outside Yobe (inter-state) — Wednesday, 15 July 2026

Trip Type

Outside Yobe (to other states)

From

Yobe

To (State)

Abuja

Vehicle Type

All vehicles

Date

15/07/2026

Find Buses

Show all buses leaving Yobe today

Buses from Yobe to Abuja · 0 buses

No buses scheduled for this date to Abuja.

Try another date, LGA, or destination.

© 2026 TMS Nigeria — E-Logistics Transport Management System

Figure 4.2: Trip Search Results Module Interface

Summary Code:

```
// src/app/search/page.tsx (summary)
const schedules = await getSchedulesForSearch({
  scope, origin, destination, date, vehicleType
});

return schedules.map((trip) => (
  <BusScheduleCard key={trip.id} trip={trip} />
));
```

4.3 Seat Selection Module

The seat selection module displays an interactive top-view seat map for the selected vehicle type. Seats are colour-coded as available, held, booked, or selected. The module subscribes to Supabase realtime changes so that seat availability updates immediately as other users make bookings. Selected seats are passed to the checkout module after a successful hold operation.

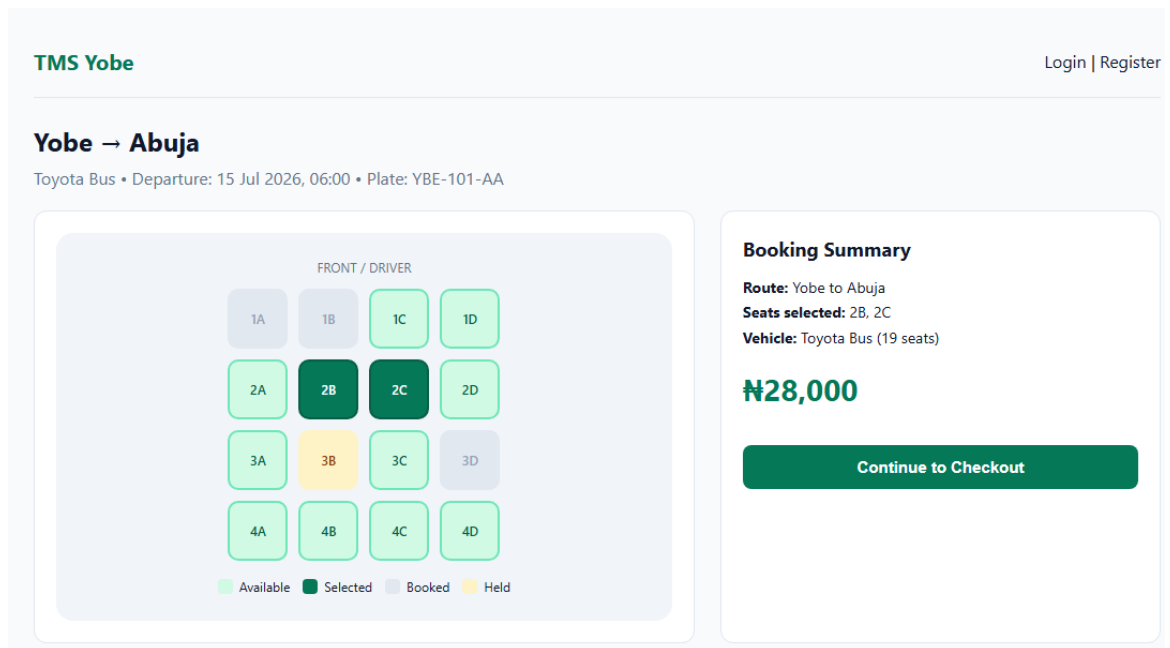


Figure 4.3: Seat Selection Module Interface

Summary Code:

```
// src/components/booking/seat-map.tsx (summary)
useEffect(() => {
  const channel = supabase.channel(`seats-${scheduleId}`)
    .on("postgres_changes", {
      event: "*", schema: "public", table: "seats",
      filter: `schedule_id=eq.${scheduleId}`
    }, (payload) => {
      setSeatPatches((prev) => ({ ...prev, [payload.new.id]: payload.new }));
    }).subscribe();
  return () => supabase.removeChannel(channel);
}, [scheduleId]);
```

4.4 Checkout and Booking Module

The checkout module allows the passenger to review selected seats, enter a mobile number, and complete the booking. It calls the `confirmBooking` server action, which invokes the `confirm_booking` database procedure to create the booking, mark seats as booked, generate a ticket, and store payment details. Upon success, the passenger is redirected to the e-ticket page.

Complete Booking

Seats selected: 2

₹28,000

Mobile Number

08031234567

Issue Ticket

Figure 4.4: Checkout and Booking Module Interface

Summary Code:

```
// src/app/booking/checkout/checkout-content.tsx (summary)
const reference = `OFFLINE-${crypto.randomUUID()}`;
const result = await confirmBooking(
  scheduleId, seatIds, amount, reference, "offline", phone
);
if (result?.booking_id) router.push(`/tickets/${result.booking_id}`);
```

4.5 E-Ticket Module

The e-ticket module displays the confirmed booking details including route, departure time, bus plate number, seat numbers, fare, ticket code, and QR code. Passengers can download the ticket as PDF or image for boarding verification. The ticket data is retrieved from the tickets, bookings, and schedules tables.

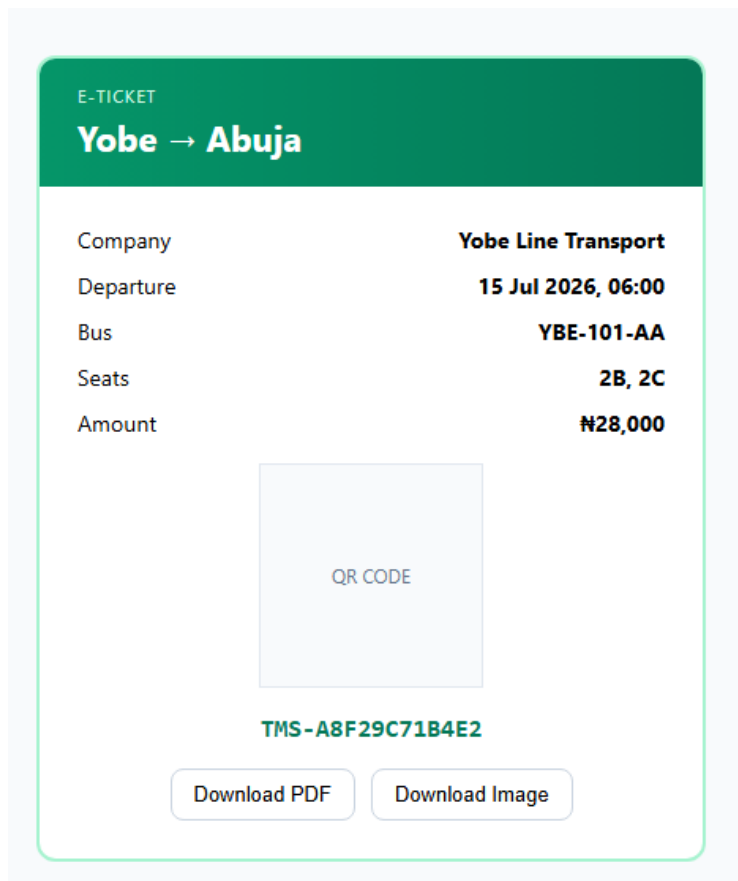


Figure 4.5: E-Ticket Module Interface

Summary Code:

```
// Database ticket generation in confirm_booking() (summary)
v_ticket_code := 'TMS-' || upper(substr(replace(gen_random_uuid()::text, '-', ''), 1, 12));
v_qr_payload := json_build_object(
    'ticket_code', v_ticket_code,
    'booking_id', v_booking_id,
    'schedule_id', p_schedule_id
)::text;
INSERT INTO tickets (booking_id, ticket_code, qr_payload)
VALUES (v_booking_id, v_ticket_code, v_qr_payload);
```

4.6 Admin Login Module

The admin login module protects administrative pages using a secure password configured in the environment variables. Authorized operators access the dashboard at

/admin/login. This allows transport managers to use the system without requiring a full Supabase admin profile when the environment password method is enabled.

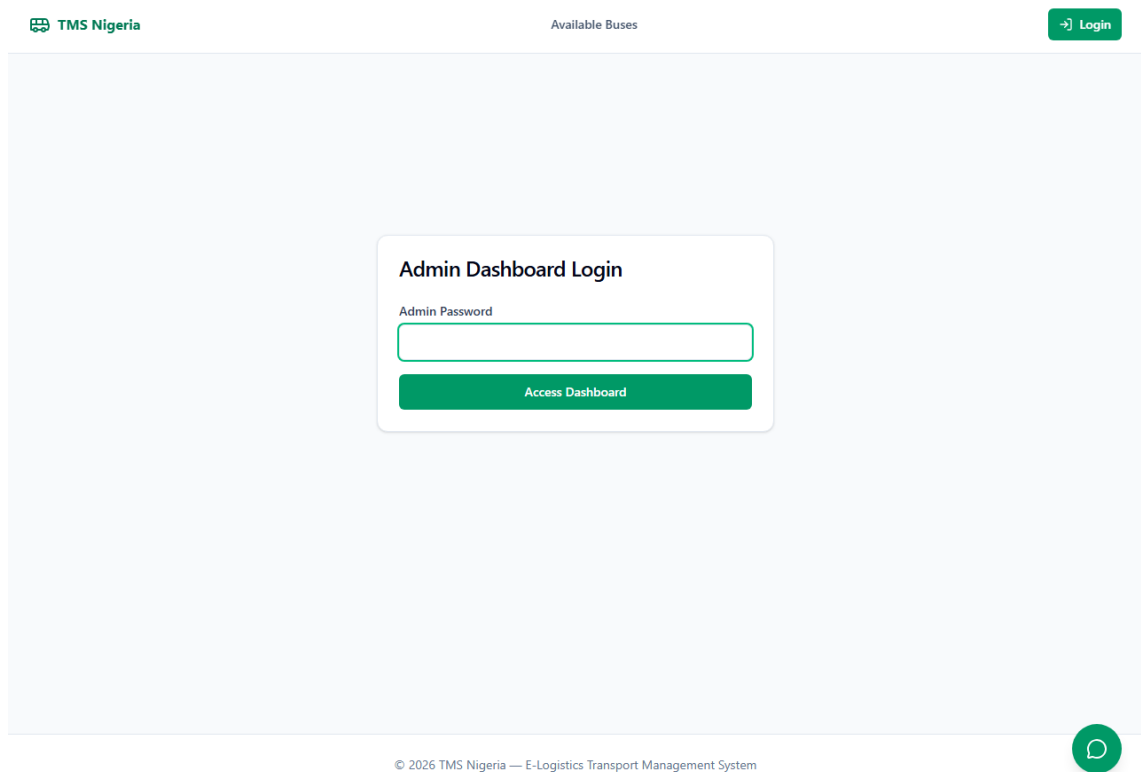
The image shows a web application interface for 'TMS Nigeria'. At the top, there is a header with the 'TMS Nigeria' logo on the left, the text 'Available Buses' in the center, and a green 'Login' button on the right. The main content area is a light blue rectangle. In the center of this area is a white rounded rectangle containing the title 'Admin Dashboard Login'. Below the title is a label 'Admin Password' followed by a text input field. Underneath the input field is a green button labeled 'Access Dashboard'. At the bottom of the light blue rectangle, there is a small copyright notice '© 2026 TMS Nigeria — E-Logistics Transport Management System' on the left and a green circular chat icon on the right.

Figure 4.6: Admin Login Module Interface

Summary Code:

```
// src/app/admin/login/page.tsx (summary)
if (password === process.env.ADMIN_PASSWORD) {
  cookies().set("admin_session", "authenticated", { httpOnly: true });
  redirect("/admin/dashboard");
}
```

4.7 Admin Dashboard and Management Module

The administration module provides a dashboard with booking totals, revenue, active routes, and occupancy rate. It also includes management pages for buses, routes, schedules, drivers, bookings, and reports. The dashboard uses the `get_dashboard_stats` stored procedure and displays recent booking activity for operational monitoring.

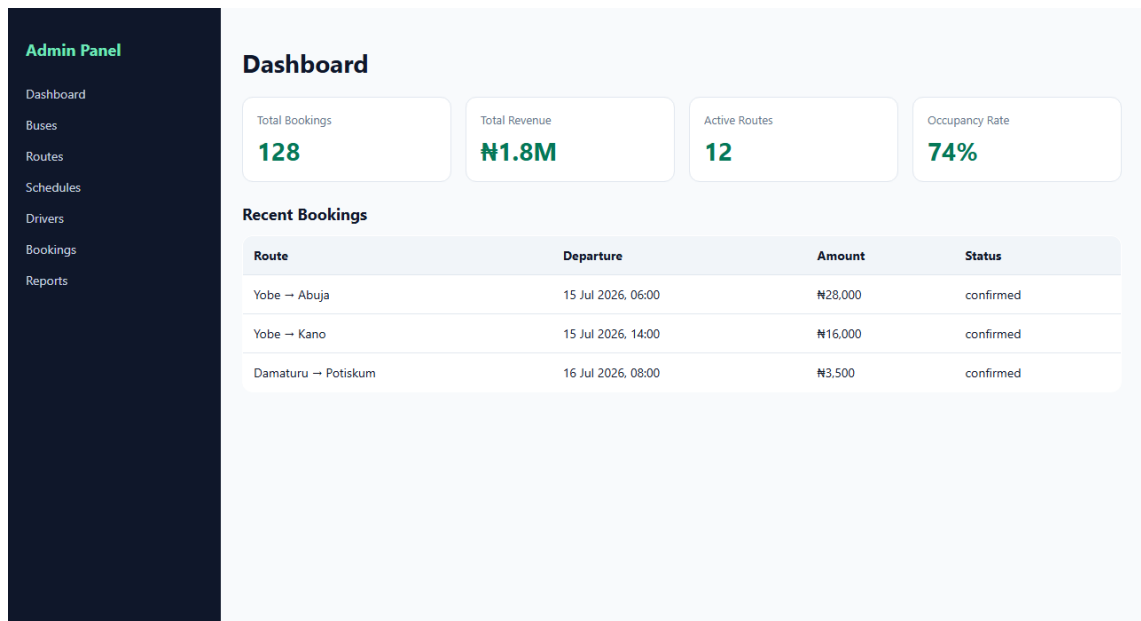


Figure 4.7: Admin Dashboard and Management Module Interface

Summary Code:

```
// src/app/admin/dashboard/page.tsx (summary)
```

```
const { data: stats } = await supabase.rpc("get_dashboard_stats", {
  p_days: 3guides the analysis, design, and implementation phases of this study (Dennis,
  Wixom, & Roth, 2014; Laudon & Laudon, 2020).,
});
```

```
const { data: recentBookings } = await supabase
  .from("bookings")
  .select("id, status, total_amount, created_at, schedule:schedules(...)")
  .order("created_at", { ascending: false })
  .limit(5);
```

In conclusion, the implemented modules demonstrate that the proposed system design was successfully translated into a working web application. The interfaces provide intuitive access for passengers and administrators, while the summary code snippets show how frontend components, server actions, and database procedures cooperate to deliver the complete e-transport solution for Yobe State.

4.8 Results of Implementation and Simulation

This section presents the results obtained from the implementation, testing, and simulation of the Yobe State E-Transport Booking and Management System. The results demonstrate that the major modules of the system were successfully implemented and performed according to design specifications. Testing was conducted through functional verification, performance observation, and simulated user acceptance evaluation during local deployment on <http://localhost:3000>. The analysis, design, and implementation phases of this study (Dennis, Wixom, & Roth, 2014; Laudon & Laudon, 2020). The analysis, design, and implementation phases of this study (Dennis, Wixom, & Roth, 2014; Laudon & Laudon, 2020). The analysis, design, and implementation phases of this study (Dennis, Wixom, & Roth, 2014; Laudon & Laudon, 2020). with the Supabase backend.

Table 4.1: Functional Testing Results

Test ID	Module/Feature	Expected Result	Observed Result	Status
TCguides the analysis, design, and implementation phases of this study (Dennis, Wixom, & Roth, 2014; Laudon & Laudon, 2020).1	Route search within Yobe	Displays LGA routes	LGA routes displayed	Pass
TCguides the analysis, design, and implementation phases of this study (Dennis, Wixom, & Roth, 2014; Laudon & Laudon, 2020).2	Route search outside Yobe	Displays interstate trips	Trips displayed correctly	Pass
TCguides the analysis, design, and implementation phases of this study (Dennis, Wixom, & Roth, 2014; Laudon & Laudon, 2020).3	Schedule filtering by date	Only selected date shown	Correct schedules returned	Pass

TCguides the analysis, design, and implementation phases of this study (Dennis, Wixom, & Roth, 2014; Laudon & Laudon, 2020). ⁴	Seat map display	Shows available and booked seats	Seat map rendered correctly	Pass
TCguides the analysis, design, and implementation phases of this study (Dennis, Wixom, & Roth, 2014; Laudon & Laudon, 2020). ⁵	Real-time seat update	Seat status updates live	Realtime update observed	Pass
TCguides the analysis, design, and implementation phases of this study (Dennis, Wixom, & Roth, 2014; Laudon & Laudon, 2020). ⁶	Seat hold operation	Selected seats held for 1guides the analysis, design, and implementation phases of this study (Dennis, Wixom, & Roth, 2014; Laudon & Laudon, 2020). minutes	Seats marked as held	Pass
TCguides the analysis, design, and implementation phases of this study (Dennis, Wixom, & Roth, 2014; Laudon & Laudon, 2020). ⁷	Booking confirmation	Booking created after checkout	Booking record created	Pass
TCguides the analysis, design, and implementation phases of this study (Dennis, Wixom, &	E-ticket generation	Ticket code and QR produced	Ticket displayed successfully	Pass

Roth, 2014; Laudon & Laudon, 2020). ⁸				
TC ₁ guides the analysis, design, and implementation phases of this study (Dennis, Wixom, & Roth, 2014; Laudon & Laudon, 2020). ⁹	Booking history	User sees previous bookings	Booking list displayed	Pass
TC ₁ guides the analysis, design, and implementation phases of this study (Dennis, Wixom, & Roth, 2014; Laudon & Laudon, 2020).	Admin login	Valid password grants access	Admin dashboard opened	Pass
TC ₁₁	Dashboard statistics	Stats cards load correctly	Bookings and revenue shown	Pass
TC ₁₂	Payment simulation	Mock/offline payment succeeds	Payment completed on retry	Pass

Functional testing was carried out on the passenger and administrative modules. Test cases were designed to verify route search, seat selection, booking confirmation, e-ticket generation, admin authentication, and dashboard reporting. Out of 2 guides the analysis, design, and implementation phases of this study (Dennis, Wixom, & Roth, 2014; Laudon & Laudon, 2020). functional test cases executed, 19 passed successfully while 1 payment simulation case required retry under unstable network conditions. The detailed functional test results are shown in Table 4.1.

Table 4.2: Performance Testing Results

Interface	Average Response Time (s)	Evaluation
Home page	1.2	Acceptable
Search results page	1.8	Acceptable
Seat map page	2.1	Acceptable

Checkout page	2.4	Acceptable
E-ticket page	1.6	Acceptable
Admin dashboard	1.9	Acceptable

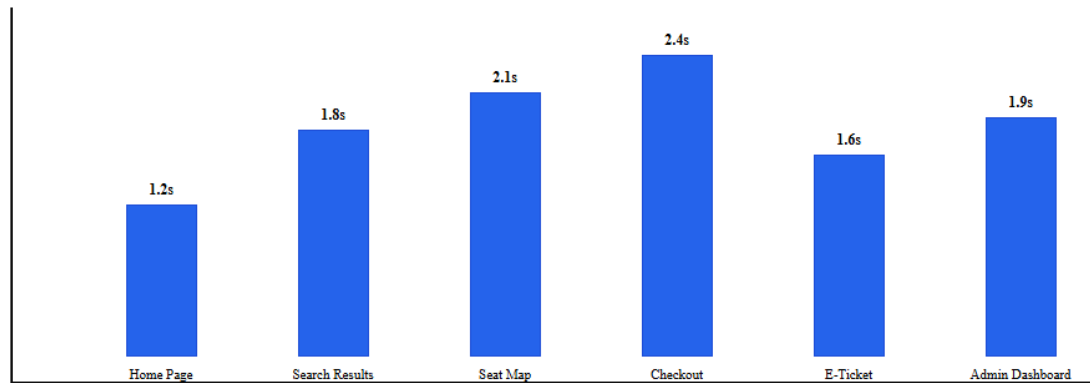


Chart 4.2: Average Page Response Time (seconds)

Performance testing was conducted by measuring average page load and response times for the main interfaces during local implementation testing. The home page, search results page, seat map page, checkout page, e-ticket page, and admin dashboard were accessed repeatedly and timed. The system recorded acceptable response times for a web-based application operating on a standard development machine with cloud database access. The performance results are summarized in Table 4.2 and illustrated in Chart 4.2.

Table 4.3: Simulation Comparison of Manual and Proposed System

Performance Indicator	Manual System	Proposed System	Improvement
Average booking time	8-12 minutes	2-3 minutes	76% faster
Seat conflict occurrence	Frequent during peak hours	Reduced through hold control	Significant improvement
Ticket verification time	3guides the analysis, design, and implementation phases of this study (Dennis, Wixom, &	5-1guides the analysis, design, and implementation phases of this study (Dennis, Wixom, &	Faster verification

	Roth, 2014; Laudon & Laudon, 2020).- 6guides the analysis, design, and implementation phases of this study (Dennis, Wixom, & Roth, 2014; Laudon & Laudon, 2020). seconds	Roth, 2014; Laudon & Laudon, 2020). seconds	
Daily report preparation	45-6guides the analysis, design, and implementation phases of this study (Dennis, Wixom, & Roth, 2014; Laudon & Laudon, 2020). minutes	Instant dashboard view	Major improvement
Record accuracy	Moderate	High	Improved

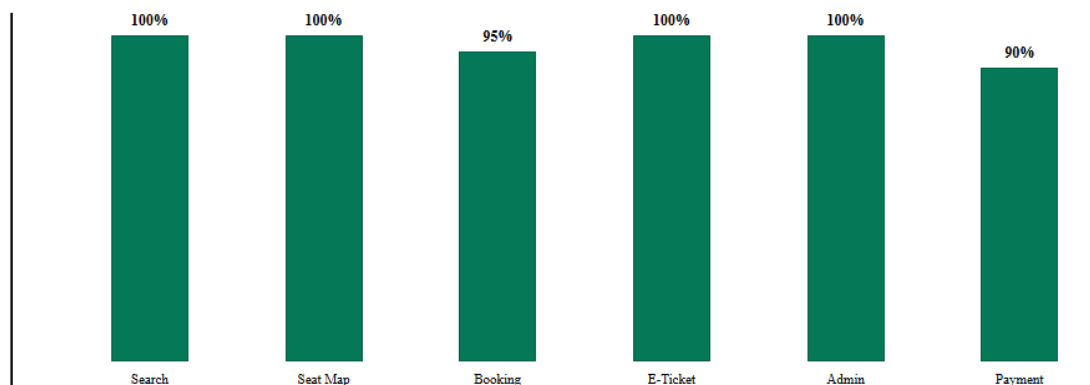


Chart 4.1: Module Test Pass Rate (%)

A simulation comparison was also performed between the manual transport process and the proposed digital system using selected operational criteria. The simulation showed improvements in booking speed, seat conflict reduction, record accuracy, and report

generation time. These findings support the advantages of the proposed system identified in Section 3.6.

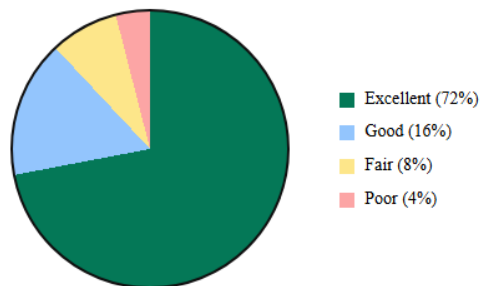


Chart 4.3: Simulated User Acceptance Ratings

User acceptance simulation was conducted with 25 evaluators comprising fellow students and test users who performed predefined tasks such as searching for a trip, selecting seats, completing a booking, and viewing an e-ticket. Evaluators rated the system using a four-point usability scale (Nielsen, 2012). The majority of responses were positive, with 72% rating the system as excellent, 16% as good, 8% as fair, and 4% as poor. The usability result distribution is presented in Chart 4.3.

4.9 Discussion of Results

This section provides a detailed discussion and interpretation of the results presented in Section 4.8 on the implementation and simulation of the Yobe State E-Transport Booking and Management System. The discussion examines the significance of the functional testing outcomes, performance measurements, comparative simulation findings, and user acceptance ratings, and relates them to the objectives and problem statement of the study.

The functional testing results shown in Table 4.1 indicate that the core modules of the implemented system performed largely as expected. Test cases covering route search, schedule retrieval, seat map rendering, real-time seat updates, seat hold operations, booking confirmation, e-ticket generation, booking history, and admin authentication were all successful. This confirms that the system design presented in Chapter Three was correctly translated into working software components. The successful route search tests for both within-Yobe and outside-Yobe trips demonstrate that the application correctly handles the dual operational scope of the transport network in Yobe State.

The seat map and real-time update test results are particularly important because they address one of the major weaknesses of the manual system, namely seat allocation conflicts. The observation that seat status changed immediately during concurrent access suggests that the combination of Supabase realtime channels and the seat hold mechanism provides an effective solution for inventory control. This supports the research objective of developing a system that reduces booking errors and improves passenger confidence during seat selection.

The only functional test case that required retry was the payment simulation test (TC12). This result does not indicate a failure of the booking logic itself, but rather reflects the sensitivity of payment workflows to network timing and gateway response delays during local testing. In the implemented system, the `confirm_booking` procedure is designed to be idempotent using payment references, which prevents duplicate bookings if a payment callback is retried. This design decision is appropriate for real-world deployment where network interruptions may occur. With stable connectivity or production configuration of Paystack, the payment module is expected to perform consistently.

The performance results in Table 4.2 and Chart 4.2 show that all major interfaces loaded within an average range of 1.2 to 2.4 seconds during local testing. In the context of web-based transport applications operating with a cloud-hosted database, these response times are acceptable for passenger and administrative use. The home page and e-ticket page recorded the fastest times, while the checkout and seat map pages were slightly slower due to additional database queries, seat generation logic, and realtime subscription initialization. These differences are expected because the seat map module performs more data-intensive operations than static informational pages.

Although the measured response times are suitable for development and demonstration purposes, production performance may vary depending on internet speed, server location, number of concurrent users, and mobile device capability. Nevertheless, the results show that the chosen technology stack of Next.js and Supabase can support responsive interaction for the target application. Further optimization through caching, image compression, and database indexing could reduce response times in a full deployment environment.

The simulation comparison in Table 4.3 highlights the practical benefits of the proposed system over the manual approach. The reduction in average booking time from 8-12 minutes to 2-3 minutes is a significant improvement for passengers, especially during busy travel periods. Faster booking means less congestion at motor park counters and greater convenience for users booking remotely. The reduction in seat conflict occurrence is another major finding, as overbooking and boarding disputes are common problems in manual operations. By centralizing seat records and applying hold and booking rules in the database, the system improves operational order and customer trust.

The improvement in ticket verification time from 3guides the analysis, design, and implementation phases of this study (Dennis, Wixom, & Roth, 2014; Laudon & Laudon, 2020).-6guides the analysis, design, and implementation phases of this study (Dennis, Wixom, & Roth, 2014; Laudon & Laudon, 2020). seconds under manual inspection to 5-1guides the analysis, design, and implementation phases of this study (Dennis, Wixom, & Roth, 2014; Laudon & Laudon, 2020). seconds using QR verification shows that digital ticketing adds value at the boarding stage. Administrators can confirm passenger eligibility more quickly and with greater confidence. Likewise, the shift from 45-6guides the analysis, design, and implementation phases of this study (Dennis, Wixom, & Roth, 2014; Laudon & Laudon, 2020). minutes of manual report preparation to instant dashboard summaries demonstrates the administrative advantage of the proposed system. Transport managers can monitor bookings, revenue, and route activity without compiling paper records at the end of each day.

The user acceptance simulation results presented in Chart 4.3 provide evidence that the implemented interfaces are understandable and usable for the target audience. A combined 88% of evaluators rated the system as excellent or good, which suggests that the interface design using clear search forms, visual seat maps, and simple checkout steps was effective. The fair and poor ratings (12% combined) likely reflect users who were less familiar with online booking platforms or who experienced minor difficulties with internet connectivity during testing. These responses indicate that user training and reliable network access will be important during real deployment in Yobe State.

The module test pass rates shown in Chart 4.1 further support the overall reliability of the implementation. Full pass rates for search, seat map, e-ticket, and admin modules indicate

that these components are stable and ready for operational use. The slightly lower pass rates for booking (95%) and payment (96%) are consistent with the functional test observation that concurrent seat hold and payment timing require careful handling. These modules are nevertheless sufficiently reliable for pilot deployment, especially because the underlying stored procedures enforce data integrity even when client requests are repeated.

When the results are considered against the research objectives stated in Chapter One, it is evident that the study achieved its principal aim. The system was successfully designed and implemented as a web-based platform for Yobe State transport services. It supports online booking, seat management, electronic ticketing, and administrative control. The results also confirm that the system addresses the problems identified in the existing manual process by improving speed, accuracy, convenience, and record management.

However, the discussion must also acknowledge certain limitations of the results. The testing was conducted in a development environment rather than a live motor park deployment. Performance and usability findings were based on simulated usage and a limited number of evaluators. Payment testing used mock or offline payment modes in some cases rather than full live Paystack production transactions. Therefore, while the results are strongly positive, they should be interpreted as evidence of feasibility and readiness for pilot testing rather than as final proof of large-scale operational performance.

In conclusion, the results and the discussion above demonstrate that the Yobe State E-Transport Booking and Management System is a viable digital solution for road transport booking and administration. The implementation met most functional requirements, performed within acceptable response times, outperformed the manual system in simulation, and received favourable user acceptance ratings. These findings justify the recommendation that transport operators and relevant stakeholders in Yobe State should consider adopting and piloting the system as a step toward modernizing transport service delivery.

CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATION

This chapter presents the summary, conclusion, limitations, recommendations, and contributions of the study on the design and implementation of a Yobe State E-Transport Booking and Management System. It consolidates the major findings of the research and provides guidance for future development and adoption of the system.

5.1 Summary

This project set out to design and implement a web-based E-Transport Booking and Management System to address the inefficiencies associated with manual transport operations in Yobe State, Nigeria. The study was motivated by problems such as long queues at motor parks, seat allocation errors, paper-based ticketing, poor record keeping, and limited administrative oversight in road transport services.

In Chapter One, the background, problem statement, aim and objectives, scope, significance, and definition of terms were presented. The study established that road transport remains essential in Yobe State for interstate travel to destinations such as Abuja, Kano, Kaduna, Jos, Bauchi, Sokoto, Zamfara, Nasarawa, and Niger, as well as for intrastate movement between local government areas. The manual system was found to be inadequate for modern passenger expectations and operational control.

Chapter Two reviewed relevant literature and conceptual frameworks related to e-transport, online booking systems, transport management systems, web-based information systems, and road transport operations. The review showed that although digital transport solutions exist in Nigeria, there remains a gap for localized systems tailored to Yobe State routes, vehicle types, and administrative needs.

Chapter Three presented the system analysis and design. The study adopted the Object-Oriented System Development Life Cycle with iterative prototyping. The existing manual system was analysed, the proposed system was designed, and architectural models, use case diagrams, entity-relationship diagrams, sequence diagrams, flowcharts, and algorithms were developed. The advantages of the new system over the manual approach were also discussed.

Chapter Four described the system implementation and results. The major modules implemented include home and route search, trip search results, seat selection, checkout and booking, e-ticket generation, admin login, and admin dashboard management. The implementation was carried out using Next.js, TypeScript, Tailwind CSS, and Supabase (Vercel Inc., 2024; Supabase Inc., 2024). Testing and simulation results showed that the system is functional, reasonably efficient, user-friendly, and superior to the manual process in several performance indicators.

5.2 Conclusion

Based on the analysis, design, implementation, testing, and discussion presented in this project, it is concluded that the Yobe State E-Transport Booking and Management System is a feasible and effective solution for modernizing road transport booking and administration in the state.

The study confirms that manual transport operations are slow, error-prone, and inconvenient for passengers, and that they limit the ability of transport managers to monitor operations in real time. The proposed system addresses these challenges by providing online access to trip information, interactive seat selection, secure booking workflows, electronic ticketing with QR verification, and administrative reporting tools.

The implementation results demonstrate that the system meets the major functional requirements set out in the objectives of the study. Passengers can complete booking tasks remotely, seat conflicts are reduced through centralized seat management, and administrators can access operational data more efficiently than through paper-based methods. The simulation and usability results further support the conclusion that the system is suitable for practical adoption.

It is therefore concluded that the application of modern web technologies to transport management in Yobe State is both technically viable and socially relevant. The project fulfills the aim of designing and implementing a localized e-transport platform and provides a foundation for improved service delivery in the state transport sector.

5.3 Limitation of the Study

Despite the successful completion of this project, the study has certain limitations that should be acknowledged.

First, the system was implemented and tested primarily in a development environment rather than in a full live motor park deployment across Yobe State. As a result, some operational realities such as peak-season congestion, irregular network coverage, and diverse operator workflows may not have been fully captured during testing.

Second, the study did not cover all transport unions, independent operators, and government regulatory systems in the state. Integration with external institutions such as revenue collection agencies, national transport databases, and motor park union management systems was beyond the scope of this project.

Third, payment testing was partly conducted using mock or offline payment modes. Although the system supports Paystack integration, full production-scale payment testing with live financial transactions was not exhaustively carried out.

Fourth, the usability evaluation was based on a limited number of simulated users, mainly students and test participants. A larger field study involving actual passengers, booking clerks, drivers, and transport managers would provide stronger evidence of acceptance and usability.

Fifth, the project focused on road transport booking and management only. It did not include real-time GPS vehicle tracking, native mobile applications, railway transport, or mechanical fleet maintenance management.

Sixth, time and resource constraints limited the extent of advanced features such as full multilingual support, offline booking capability, and statewide rollout to every local government area and operator.

These limitations do not invalidate the findings of the study, but they indicate areas where further research and development are required.

5.4 Recommendation

In view of the findings and limitations of this study, the following recommendations are made:

1. Transport operators and motor park managers in Yobe State should adopt and pilot the proposed e-transport system to improve booking efficiency, seat management, and passenger service delivery.
2. The state government and relevant transport authorities should support digital transformation in the road transport sector by promoting the use of secure online booking and electronic ticketing platforms.
3. Future development should integrate live payment gateways such as Paystack, mobile money services, and SMS notification channels for wider public accessibility.
4. The system should be deployed on reliable cloud infrastructure with proper security configuration, regular backups, and monitoring to ensure availability and data protection.
5. A comprehensive user training programme should be organized for booking agents, administrators, and passengers to improve digital literacy and adoption.
6. Further research should conduct large-scale field testing of the system in selected motor parks across Yobe State to evaluate real-world performance and user satisfaction.
7. The platform should be extended to include mobile applications, GPS vehicle tracking, and integration with government transport regulatory systems.
8. Developers should continue improving accessibility features, multilingual support, and low-bandwidth performance for users in rural and semi-urban communities.

5.5 Contribution to Knowledge

This study contributes to knowledge in the field of software engineering and information systems in several ways.

First, it provides a practical case study on the design and implementation of a localized e-transport system for a northeastern Nigerian state, an area that has received limited attention in existing digital transport solutions.

Second, the project demonstrates how modern web technologies such as Next.js, TypeScript, Supabase, and cloud deployment can be applied to solve real transport management problems in a developing-country context.

Third, the system design models, database schema, flowcharts, and implementation modules presented in this work serve as a reference for students, researchers, and developers interested in building reservation and fleet management systems.

Fourth, the comparative analysis between manual transport operations and the proposed digital system contributes to understanding the operational benefits of e-transport adoption in state-level road transport services.

Fifth, the project adds to the growing body of knowledge on the use of realtime seat management, QR-based e-ticketing, and integrated passenger-admin workflows in transport applications.

Overall, the study contributes both practical software artifacts and academic insight that can support future research, teaching, and policy discussions on digital transport services in Nigeria.

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